Towards Faithful Data Selection: Statistical Limits and Scalable Algorithms

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Abstract

What does it mean to have an "optimal" data selection algorithm? Optimality is conventionally framed as finding a subset that minimizes the population error of the classifier trained only on that subset. Our starting point in this work is first to show that this goal is information-theoretically impossible unless the data selection algorithm has access to a validation set order-wise larger than the training set. This impossibility result motivates shifting from unverifiable population risk minimization to a one-sided guarantee that we refer to as *faithfulness*: for any candidate subset, we provide an estimable proxy that upper-bounds the difference between training on the full training data and on only the subset. We model training with the Unadjusted Langevin Algorithm and bound the Kullback–Leibler divergence between subset-trained and full-data-trained parameter distributions by the trajectory-accumulated gradient discrepancy. Minimizing this discrepancy gives a principled way to select a subset. Even though the resulting problem is NP-hard, we propose a swap-based local search that decreases the bound at every accepted step. It can therefore refine the output of any selector without worsening its bound. Finally, we provide experimental results on CIFAR10 and CIFAR100 showing that it matches existing data selection baselines. Implementation of our algorithm and the code for reproducing the results are available at: https://github.com/Mehdi-Inane/data_selection_swap.

1 Introduction

Training on ever-larger datasets is expensive, and not all samples contribute equally: careful pruning can improve how performance scales with dataset size (Sorscher et al., 2023). *Data selection* formalizes this idea: given a training set S of n samples and a learning algorithm $\mathcal{A}$, find a subset $I \subseteq S$ of size $m < n$ such that training on I alone performs as well as training on all of S. Besides reducing computation and storage, selection is a concrete instance of the data-centric view of machine learning (Zha et al., 2025), in which the dataset, rather than the model, is the object to curate.

There is a vast body of work on data selection algorithms (Moser et al., 2025), which differs in what the data selection algorithm has access to and in how it uses this information. Bi-level optimization approaches (Killamsetty et al., 2021b; Xia et al., 2024; Nikdan et al., 2026) select subsets of the training set based on the error on a held-out validation set, and training-data based methods use *only* information stemming from the training set and the learning dynamics induced by a learning algorithm (Toneva

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et al., 2019; Paul et al., 2021; Killamsetty et al., 2021a; Cohen-Addad et al., 2026). Despite this empirical progress, the statistical limits of data selection remain poorly understood.

There has also been a recent line of theoretical work on the statistical aspects of data selection. For instance, data memorization is inherently related to data selection, since memorization identifies training examples that a learner cannot afford to discard. The work of Feldman (2020) shows theoretically that on long-tailed distributions, rare examples may provide the only information about subpopulations that collectively carry substantial test mass, making their memorization necessary for near-optimal accuracy. Empirically, Feldman and Zhang (2020) show that on standard image benchmarks memorized examples often improve accuracy on related test points, so dropping rare samples may degrade the test accuracy. In recent work, Hanneke et al. (2025) study the following fundamental question: for empirical risk minimizers, given a set of training samples, does there exist a (smaller) *good* subset such that training on it performs nearly as well as training on all the samples? Their main result is to give affirmative answers in various settings. Another line of work (Kolossov et al., 2023) identifies settings in which selection improves over full-sample training, and provides settings where popular selection methods like unbiased reweighting and influence-based rules are suboptimal. Neither work, however, asks what a selector can identify from the samples alone. In this work, we complete this picture by asking: when the training sample contains a subset on which training yields an accurate model, what can a data selection algorithm infer about this subset from finite training (and possibly validation) samples?

To make this question precise, let $S = (Z_1, \ldots, Z_n)$ be i.i.d. samples from an unknown distribution P, and let $\mathcal{A}$ be a learning algorithm. For $I \subseteq [n]$, write $R_P(\mathcal{A}(S_I))$ for the population risk under P of the model trained on $S_I = (Z_i)_{i \in I}$, and $R_P^\star$ for the Bayes error. Given a budget m, let $\widehat{I} \subseteq [n]$ with $|\widehat{I}| = m$ be the subset chosen by our data selection algorithm, and let $I^\star \in \arg\min_{I \subseteq [n],\, |I| = m} R_P(\mathcal{A}(S_I))$ be the best subset with the same budget. Writing $\mathcal{A}(S_{\widehat{I}})$ to denote the hypothesis learned from $S_{\widehat{I}}$, we consider two ways of decomposing the excess risk, $R_P(\mathcal{A}(S_{\widehat{I}})) - R_P^\star$:

$$\underbrace{R_P(\mathcal{A}(S_{I^\star})) - R_P^\star}_{\text{compression error}} + \underbrace{R_P(\mathcal{A}(S_{\widehat{I}})) - R_P(\mathcal{A}(S_{I^\star}))}_{\text{selection error}}, \tag{1}$$

$$\text{and} \quad \underbrace{R_P(\mathcal{A}(S)) - R_P^\star}_{\text{full-data excess risk}} + \underbrace{R_P(\mathcal{A}(S_{\widehat{I}})) - R_P(\mathcal{A}(S))}_{\text{faithfulness gap}}. \tag{2}$$

The first decomposition separates two questions: whether the realized training sample contains a good subset of size at most m (a question that Hanneke et al. (2025) studies), and whether the data selection algorithm can identify one. In our statistical setting, however, the comparator $\mathcal{A}(S_{I^\star})$ is not available to the data selection algorithm, because $I^\star$ depends on the unknown distribution P. The second decomposition replaces this inaccessible comparator with the full-data model $\mathcal{A}(S)$; both $\mathcal{A}(S_{\widehat{I}})$ and $\mathcal{A}(S)$ can be trained on the observed data. This motivates two questions: **(1)** When one or more good subsets of size m exist, can a data selection algorithm find a good subset, and do so using only the training sample and a validation set of moderate size? **(2)** When this is impossible, can the algorithm use the training data alone to ensure faithfulness to full-data training, and can it optimize this faithfulness at scale?

CONTRIBUTIONS

- **A statistical limit on the selection error.** We separate the existence of a good subset from the ability of a data selector to find it. For every compression fraction $\lambda \in (0, 1)$, training-set size n, and validation-set size v, we construct a finite family $\mathfrak{P}_{\lambda,n,v}$ of zero-Bayes-error classification problems with two properties (Theorem 2.1). First, the family is not degenerate: there is a learner $\mathcal{A}^*$ that, on every problem in the family, attains vanishing excess risk both when trained on all n examples and when trained on its best subset of size $m = \lfloor \lambda n \rfloor$, so an accurate subset always exists. Second, no

selector can find one: for every learner $\mathcal{A}$ and every selector Π that sees only the training sample and a validation set of size v, some problem in the family forces

$$\mathbb{E}\big[\underbrace{R_P(\mathcal{A}(S_{I^\star})) - R_P^\star}_{\text{compression error}} + \underbrace{R_P(\mathcal{A}(S_{\widehat{I}})) - R_P(\mathcal{A}(S_{I^\star}))}_{\text{selection error}}\big] \geq \Omega_\lambda\left(\frac{n}{n+v}\right).$$

Hence every learner whose compression error vanishes, $\mathcal{A}^*$ among them, pays a selection error of order $n/(n+v)$ with every selector: a validation set of $v/n \to \infty$ is required to make it vanish.

- **Faithfulness, with a training-data only proxy.** The negative result points to the second decomposition: we ask for a *faithful* subset, one whose training approximates the same distribution over models as training on S. By modeling training dynamics of gradient-based methods with the Unadjusted Langevin Algorithm (ULA), we bound the faithfulness by the trajectory-accumulated gradient discrepancy $G_T(I)$ (Proposition 4.1). This gives a *faithfulness proxy* for any candidate subset. In Figure 1 we show that the trajectory gradient discrepancy G_T correlates with the faithfulness gap for SGD-trained models.
- **A simple algorithm with an improvement guarantee.** Minimizing $G_T(I)$ is NP-hard, and we do not claim to solve it optimally. Existing gradient-based selectors only optimize relaxations of it, for instance by matching gradients at a single training iterate or by scoring examples independently (Remark 4.3), so none directly target the quantity our bound controls. Instead, we give a simple swap-based local search algorithm that, starting from any subset, halts or identifies a subset I that decreases $G_T(I)$. One can start from the full training data, or aim to refine the output of another selector (Theorem 4.6). Its cost is dominated by per-example gradients, which we approximate using a single early checkpoint, yielding an estimate $\widehat{G}_T$ of G_T. On CIFAR-10 and CIFAR-100, this simple method is on par with existing static data selection baselines. This opens up avenues for improving the optimization of $G_T(I)$ as a way to improve data selection.

2 Statistical Limits of Optimal Subset Selection

In this section, we start by defining the task of finding the optimal subset in the training set. A natural comparator (or goal) is as follows: Fix a possibly randomized learning algorithm $\mathcal{A} = \{\mathcal{A}_n\}_{n\in\mathbb{N}}$, where $\mathcal{A}_m$ is the learner applied to samples of size m and a training sample $S = (Z_1, \ldots, Z_n) \sim P^n$. Assume that we have a budget of size m for data selection. For every index set $I \subseteq [n]$, let $S_I = (Z_i)_{i\in I}$, with the indices in increasing order. Let

$$I_m^\star \in \operatorname*{arg\,min}_{\substack{I\subseteq[n]\\|I|=m}} R_P(\mathcal{A}_m(S_I)), \tag{3}$$

with ties broken canonically. Thus $R_P(\mathcal{A}_m(S_{I_m^\star}))$ is the risk of the best subset that an oracle with access to P can find in the realized sample. Importantly, the oracle must use the same learner $\mathcal{A}$ as the data selector.

Let $V \sim P^v$ be an independent validation set, and let $\Pi : \mathcal{Z}^n \times \mathcal{Z}^v \longrightarrow \{I \subseteq [n] : |I| = m\}$ be a (possibly randomized) data selection algorithm. Write $\widehat{I}_m := \Pi(S, V)$ for the selected subset. Letting $\mathcal{A} = \mathcal{A}_m$ denote the learning algorithm fed m samples, the excess risk of the hypothesis $\mathcal{A}(S_{\widehat{I}_m})$ can be decomposed into the compression error and selection error as in Equation (1). Both terms in this decomposition are nonnegative: the first asks whether the training sample contains a good size-m subset for $\mathcal{A}_m$, and the second measures the cost of finding one using Π. As we will show, there exists a class of problems for which we have a dichotomy: every size-m subset has large compression error for $\mathcal{A}_m$, or a good subset exists, but Π cannot identify it.

Theorem 2.1. *Consider multiclass classification with ordinary 0–1 loss. Fix any compression fraction $\lambda \in (0,1)$, and set $K := 2^{\lceil 100/(1-\lambda)\rceil}$. Let the label space be $[K]$. There exist $n_\lambda \in \mathbb{N}$ and $C_\lambda > 0$, depending only on λ, with the following property: For every integer $n \geq n_\lambda$ and integer validation size $v \geq 0$, setting $m = \lfloor \lambda n \rfloor$, there is a finite family $\mathfrak{P}_{\lambda,n,v}$ of deterministic-label K-class classification distributions, together with an explicit deterministic learner $\mathcal{A}^*$, such that, (1) every member of the family has zero Bayes risk and the risk of the model learned by $\mathcal{A}^*$ converges to zero; (2) for every pair consisting of a learner $\mathcal{A}_m$ and a data selector Π, there exists $P \in \mathfrak{P}_{\lambda,n,v}$ such that, for independent samples $S \sim P^n$ and $V \sim P^v$,*

$$\begin{aligned} &\mathbb{E}_S\big[R_P(\mathcal{A}_m(S_{I_m^\star})) - R_P^\star\big] \\ &+ \mathbb{E}_{S,V,\Pi}\Big[R_P\Big(\mathcal{A}_m(S_{\widehat{I}_m})\Big) - R_P(\mathcal{A}_m(S_{I_m^\star}))\Big] \\ &\geq \frac{\lambda(1-\lambda)^2}{10^6}\frac{n}{n+v}, \end{aligned} \tag{4}$$

where $\widehat{I}_m = \Pi(S,V)$ satisfies $|\widehat{I}_m| = m$, and $I_m^\star$ minimizes $R_P(\mathcal{A}_m(S_I))$ over $I \subseteq [n]$ with $|I| = m$. Moreover, these families and the corresponding learners can be chosen so that, for every $n \geq n_\lambda$ and for every $P \in \mathfrak{P}_{\lambda,n,v}$

$$\frac{n+v}{n}\,\mathbb{E}_S[R_P(\mathcal{A}^*(S)) - R_P^\star] \leq \frac{C_\lambda}{\log n}, \tag{5}$$

$$\frac{n+v}{n}\,\mathbb{E}_S[R_P(\mathcal{A}^*(S_{I_m^\star})) - R_P^\star] \leq \frac{C_\lambda}{\log n}. \tag{6}$$

In Equation (6), $I_m^\star$ is the optimal subset for $\mathcal{A}^*$.

The full proof is given in Section B. Let us unpack the implications of this result:

Accurate subsets exist: Bayes risk is zero, and there exists an algorithm for the class of problems ($\mathcal{A}^*$ in Theorem 2.1) that achieves vanishing error on both the full sample and a distribution-aware oracle subset of at most $\lfloor \lambda n \rfloor$ examples. Thus the problem is non-degenerate.

Sample-based selection is hard: Finding such a subset can nevertheless be statistically hard: for fixed λ and $v = O(n)$, for every selector-algorithm pair such that the algorithm has small compression error, the selection error is bounded below by a constant even with unlimited computation.

Tightness: Our lower bound in Theorem 2.1 is tight in its dependence on the number of training samples. Specifically, conditional on the training sample S, running $\mathcal{A}_m$ on all subsets of S of size m induces a finite hypothesis class of size at most $(en/m)^m$. Standard generalization bounds (Shalev-Shwartz and Ben-David, 2014) guarantee that a validation set of size $v = \Theta(\lambda n/\varepsilon^2)$ suffices to achieve selection error ε.

Remark 2.2 (long-tail distributions in practice)**.** *The family of distributions in our lower bound shares the long-tail structure of Feldman (2020): there are as many rare subpopulations as training examples, so many of them appear once or never in the training set. There, the subpopulations are unrelated, so one that is absent from the training set cannot be predicted, even by full-data training or by an oracle that knows the distribution. Our main technical contribution is to relate rare subpopulations to one another. This is why an accurate subset of size m exists: an oracle that knows which rare subpopulations will matter at test time keeps only the examples needed to reconstruct them.*

Proof overview of Theorem 2.1. The natural starting point is to consider a long-tailed distribution from Feldman (2020) with many rare subpopulations so that accurate prediction requires retaining a large amount of information from the training sample. The immediate difficulty is that if predicting well

requires storing more than m pieces of information, then even full-data training and a population-aware oracle with budget m will fail. Our construction resolves this difficulty through two main ideas. First, we include a shared baseline task that every learner must solve at test time; both the oracle and any data selector must spend almost all of their subset budget m on this baseline task, leaving only a small residual budget for the remaining subpopulations. Second, we connect the remaining subpopulations through a sparse graph where each subpopulation's label is determined by the labels of its immediate neighbors (using Erdős–Sachs). By connecting the tasks through a hidden graph, the oracle (as well as full-data training) can deduce the exact label of other active subpopulations. □

3 Faithful data selection

The previous section studied the conventional goal of data selection: returning a subset of the training sample whose population error nearly matches that of the best subset $I_m^\star$ in Equation (3). However, our results show that for every pair consisting of a learning algorithm and a data selection algorithm, there exists a scenario in which the data selection algorithm incurs a non-vanishing selection error unless it has $\omega(n)$ validation samples, even though a simple learner trained on all of S succeeds there. We therefore change the goal.

The selection error in Equation (1) compares the chosen subset with $I_m^\star$, which depends on the data distribution P and is unknown to the selector. Instead, let us consider comparisons to the model trained on all of S, which the selector can access. In particular, consider the decomposition of the excess risk of the selected subset described in Equation (2). The first term (full-data excess risk) does not involve the selector and is typically small in practice, since data selection is usually applied when training on the full sample already performs well. Therefore, excess risk of the selected subset is then essentially the second term: the faithfulness gap. Motivated by this, we seek a subset on which training yields a model that closely tracks the full-data model.

Remark 3.1. *We do not claim that we can overcome the fundamental lower bound in Theorem 2.1, since the result provides a lower bound on the excess error directly. Indeed, the hard class of distributions we introduced is such that the faithfulness gap is large. When the gap is small, perhaps we can find and optimize faithful subsets. As we show next, we provide a practical proxy that can be computed from the training set and has the property that, if the proxy is small, we know that the faithfulness gap is small.*

3.1 From faithfulness to KL divergence

How can we control the faithfulness gap? The faithfulness gap in Equation (2) is the difference between the averages of one function, $\theta \mapsto R_P(\theta)$, under two distributions over trained parameters: the law $\mathcal{A}_n(S)$ of the model trained on S and the law $\mathcal{A}_m(S_{\widehat{I}})$ of the model trained on the subset. We cannot evaluate this function, since P is unknown, but we do not need to: if the two *laws* are close, every bounded function of θ has nearly the same average under both, whatever P is. We therefore call a subset *faithful* if training on it produces nearly the same distribution over models as training on S. This requirement is stronger than a small faithfulness gap, and usefully so: it controls the gap under every data distribution at once, and it controls every other bounded evaluation of the trained model, which we call a *query*. Examples include accuracy on a subgroup or on a rare subpopulation, accuracy under distribution shift, and the loss or prediction at a single test point as in data attribution (Ye et al., 2024). The next result, whose proof (Section C) is simply based on the variational representation of total variation distance and Pinsker's inequality, formalizes this intuition:

Proposition 3.2. *Let $a, b \in \mathbb{R}$ and $n, m \in \mathbb{N}$ such that $1 \le m \le n$. Let $q : \mathbb{R}^d \to [a, b]$ be a measurable bounded function. Fix $S \in \mathcal{Z}^n$ as a training set of size n and let $I \subseteq [n]$ be an arbitrary subset of $[n]$*

such that $|I| = m$. *Then, for every randomized algorithm* $\mathcal{A} = \{\mathcal{A}_m\}_{m\in\mathbb{N}}$, *we have:*

$$\left|\mathbb{E}_{\theta\sim\mathcal{A}_n(S)}[q(\theta)] - \mathbb{E}_{\theta'\sim\mathcal{A}_m(S_I)}[q(\theta')]\right| \leq \sqrt{\frac{(b-a)^2}{2} D_{\mathrm{KL}}(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I))}.$$

Note that $\mathcal{A}_n$ *and* $\mathcal{A}_m$ *should both share the same initialization distribution.*

4 Practical Data Selection using Langevin Algorithm

Proposition 3.2 reduces faithfulness to a single number, $D_{\mathrm{KL}}(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I))$. How can we turn this result into a faithful data selection algorithm? To select subsets with it, we need two things: an upper bound on this divergence that can be evaluated based on the training set, and a form of that bound that can be minimized over subsets.

In this section, we focus on gradient-based learning algorithms that are often used in practice. To operationalize the KL bound, we approximate gradient-based training by the Unadjusted Langevin Algorithm (ULA). This approximation lets us bound the divergence in closed form and then turn the bound into a practical and scalable algorithm.

More precisely, assume we have a fixed loss function $L_S(\theta) = \frac{1}{|S|}\sum_{x\in S}\ell(\theta, x)$ and let d be the number of parameters. Fix a number of iterations $T \geq 1$, step sizes $h_0, \ldots, h_{T-1} > 0$, inverse temperatures $\beta_0, \ldots, \beta_{T-1} > 0$. On a training set S', ULA starts from an initialization distribution $\theta_0 \sim \mu_0$ and, for $t = 0, \ldots, T-1$, let

$$\theta_{t+1} = \theta_t - h_t \nabla L_{S'}(\theta_t) + \sqrt{2h_t/\beta_t}\,\xi_{t+1}, \tag{7}$$

where $\xi_1, \xi_2, \ldots$ are i.i.d. $\mathcal{N}(0, I_d)$ and independent of θ_0. We write $\mu_t^{S'}$ for the law of θ_t, and we take the learner to be ULA stopped at iteration T, that is, $\mathcal{A}_k(S') := \mu_T^{S'}$ for every sample S' of size k. ULA is gradient descent with injected Gaussian noise, and its stochastic-gradient version is the SGLD algorithm of Welling and Teh (2011). ULA is an approximation of how models are trained in practice. Our guarantees are proven for ULA, and Section 5 shows that the resulting selection rule remains effective when models are trained with SGD.

Before stating our results, we define a quantity that we refer to as the *trajectory gradient discrepancy*,

$$G_T(I) := \sum_{t=0}^{T-1} \frac{\beta_t h_t}{4} \mathbb{E}_{\theta_t\sim\mu_t^S} \left\|\nabla L_{S_I}(\theta_t) - \nabla L_S(\theta_t)\right\|^2, \tag{8}$$

the squared distance between the subset and full-data gradients, accumulated along the full-data training trajectory. The proof is deferred to Section D.

Proposition 4.1. *Fix* $T \geq 1$, *step sizes* $h_0, \ldots, h_{T-1} > 0$, *inverse temperatures* $\beta_0, \ldots, \beta_{T-1} > 0$, *integers* $1 \leq m \leq n$, *a training set* $S \in \mathcal{Z}^n$, *and* $I \subseteq [n]$ *with* $|I| = m$, *and assume that* $\ell(\cdot, z)$ *is continuously differentiable for every* $z \in \mathcal{Z}$. *Assume furthermore that* $\mathbb{E}_{\theta\sim\mu_t^S}\left[\|\nabla\ell(\theta, Z_i)\|^2\right] < \infty$ *for all* $i \in [n], 0 < t \leq T-1$. *Run ULA on* S *and on* S_I *with these schedules. Then,* $D_{\mathrm{KL}}(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I)) \leq G_T(I)$.

Corollary 4.2 (Faithfulness proxy under ULA)**.** *Under the assumptions of Proposition 4.1, for all* $a < b$ *and every measurable* $q : \mathbb{R}^d \to [a, b]$,

$$\left|\mathbb{E}_{\theta\sim\mathcal{A}_n(S)}[q(\theta)] - \mathbb{E}_{\theta\sim\mathcal{A}_m(S_I)}[q(\theta)]\right| \leq (b-a)\sqrt{G_T(I)/2}.$$

The path-space identity behind Proposition 4.1 is the same quantity that appears as "gradient incoherence" in information-theoretic generalization bounds for SGLD (Negrea et al., 2019; Haghifam et al., 2020), and KL divergences between Langevin chains on neighbouring datasets drive Langevin unlearning (Chien et al., 2024; Inane et al., 2026) and leave-one-out distinguishability (Ye et al., 2024).

Our contribution is not the bound but its use: we read G_T as an estimable, distribution-free *proxy* for the selection objective (we show in Section 5 that an estimate of G_T can predict the downstream test accuracy). A notable property is that the expectation in equation 8 is over the full-data chain only, so $G_T(I)$ can be estimated for every candidate subset from the same full-data training runs, using per-example gradients; this requires no training on any subset and no access to P or to held-out data. In Figure 1, we estimate G_T using a single full training run for a stochastic gradient descent (SGD) trajectory and show that it still correlates with the faithfulness error.

Remark 4.3 (G_T and existing gradient-based selectors)**.** *At a single iterate, with $g_i := \nabla\ell(\theta, Z_i)$ and $\bar{g} := \nabla L_S(\theta)$,*

$$\Big\|\tfrac{1}{m}\sum_{i\in I} g_i - \bar{g}\Big\|^2 \;=\; \tfrac{1}{m^2}\sum_{i,j\in I}\langle g_i, g_j\rangle \;-\; \tfrac{2}{m}\sum_{i\in I}\langle g_i, \bar{g}\rangle \;+\; \|\bar{g}\|^2.$$

Several selectors optimize pieces of this expansion. CRAIG (Mirzasoleiman et al., 2020) and GRAD-MATCH (Killamsetty et al., 2021a) minimize the same discrepancy at the current iterate but over weighted subsets, via a facility-location surrogate or orthogonal matching pursuit, and GRAD-MATCH re-solves it during training; G_T is the unweighted, trajectory-averaged version, and Proposition 4.1 is what connects it to the trained model. Ranking examples by alignment $\langle g_i, \bar{g}\rangle$ (TracIn-style scores with the training loss as target; Pruthi et al., 2020) optimizes the linear term alone and ignores the Gram term, i.e. redundancy among the selected examples. GraNd (Paul et al., 2021) ranks by $\|g_i\|$, which enters G_T only through the diagonal of the Gram term; since GraNd and EL2N keep the largest scores, they increase this term, which is consistent with their subsets being the least faithful ones in Figure 1.

Based on the trajectory gradient discrepancy, we define the following quadratic integer programming problem (more details in Section E):

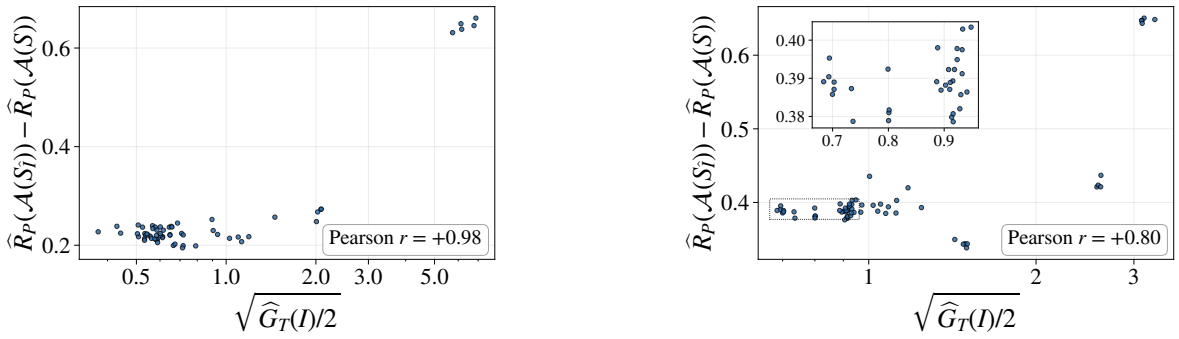
Definition 4.4 (Selection as a quadratic program)**.** *In the setting of Proposition 4.1, write $\zeta_i(\theta) := \nabla\ell(\theta, Z_i) - \nabla L_S(\theta)$ for the centered per-example gradients and define $C \in \mathbb{R}^{n\times n}$ by $C_{ij} := \sum_{t=0}^{T-1} \frac{\beta_t h_t}{4}\,\mathbb{E}_{\theta\sim\mu_t^S}\langle\zeta_i(\theta), \zeta_j(\theta)\rangle$. Then C is positive semidefinite, $C\mathbf{1} = 0$, and $G_T(I) = \frac{1}{m^2}\mathbf{1}_I^\top C\,\mathbf{1}_I$ for every $I \subseteq [n]$ with $|I| = m$. Consequently, minimizing $G_T(I)$ over subsets of size m is the integer quadratic program*

$$\min_{\alpha\in\mathcal{B}_m} D(\alpha) \triangleq \frac{1}{m^2}\alpha^\top C\alpha,$$

where $\mathcal{B}_m := \{\alpha \in \{0,1\}^n : \mathbf{1}^\top\alpha = m\}$.

4.1 Subset selection via gradient swaps

How can we solve the program in Definition 4.4 when n is large? Three facts rule out the obvious approaches. First, exact minimization of the program in Definition 4.4 is NP-hard even for $T = 1$ and $d - 1$ by reduction from fixed-cardinality subset sum (and general MIQP is NP complete; (Pia et al., 2014)). Second, forming C requires $\Theta(n^2)$ inner products at every point of the trajectory, and storing it requires $\Theta(n^2)$ memory. Third, relaxing α to $[0,1]^n$ is uninformative: since $C\mathbf{1} = 0$, the fractional point $\frac{m}{n}\mathbf{1}$ has objective value zero and says nothing about which subset to select. These observations prompt us to focus on designing an algorithm that requires only matrix–vector products with C, keeps every iterate a subset of size m, and never increases G_T. Algorithm 1 is such a method: a local search that exchanges one selected example for one unselected example at a time.



(a) ResNet-18 on CIFAR-10

(b) VGG-16 on CIFAR-100

Figure 1: Empirical Faithfulness gap $\widehat{R}_P(\mathcal{A}(S_I)) - \widehat{R}_P(\mathcal{A}(S))$ versus the estimate of $\sqrt{\widehat{G}_T(I)/2}$ for CIFAR-10 / ResNet-18 (left) and CIFAR-100 / VGG-16 (right) at retention fraction 0.1. Each point represents a subset from a data selection method, and $\mathcal{A}$ represents the SGD training procedure. Subsets with large $G_T(I)$ are the ones with the largest gap. The inset (right) zooms in on the dotted region. Points in the upper-right corner correspond to EL2N (EL2N in Figure 2 achieves lower accuracy), confirming the theoretical prediction of Corollary 4.2.

Algorithm 1 Gradient-ranked swap descent

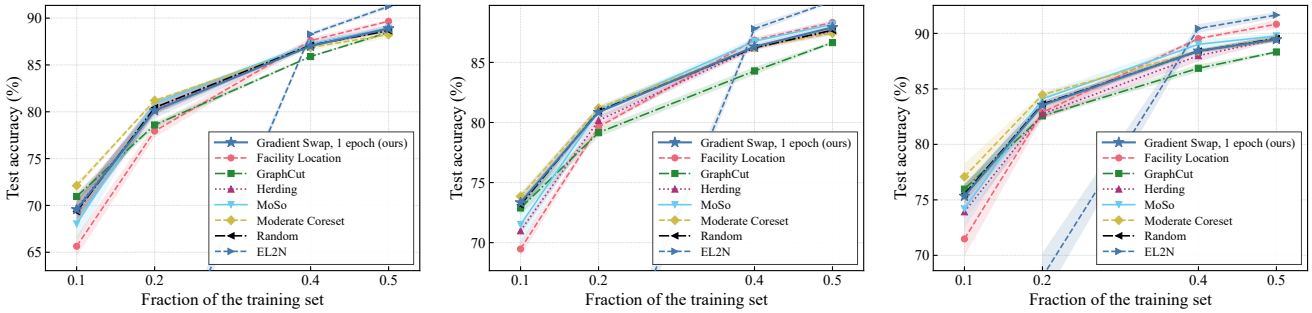
Require: products $v \mapsto Cv$; budget $1 \le m \le n-1$; initial subset $I_0 \subseteq [n]$ with $|I_0| = m$
Ensure: subset $I_{\text{out}} \subseteq [n]$ with $|I_{\text{out}}| = m$

$s_0 \leftarrow \frac{2}{m^2} C\mathbf{1}_{I_0}$ ▷ $s_0 = \nabla D(\mathbf{1}_{I_0})$
for $k = 0, 1, 2, \ldots$ **do**
  $i_k \in \arg\max_{i \in I_k} s_k(i)$ ▷ drop
  $j_k \in \arg\min_{j \notin I_k} s_k(j)$ ▷ add
  $I' \leftarrow (I_k \setminus \{i_k\}) \cup \{j_k\}$
  $s' \leftarrow \frac{2}{m^2} C\mathbf{1}_{I'}$
  if $G_T(I') < G_T(I_k)$ **then** ▷ $G_T(I) = \frac{1}{2}\mathbf{1}_I^\top s$
    $(I_{k+1}, s_{k+1}) \leftarrow (I', s')$
  else
    return $I_{\text{out}} := I_k$

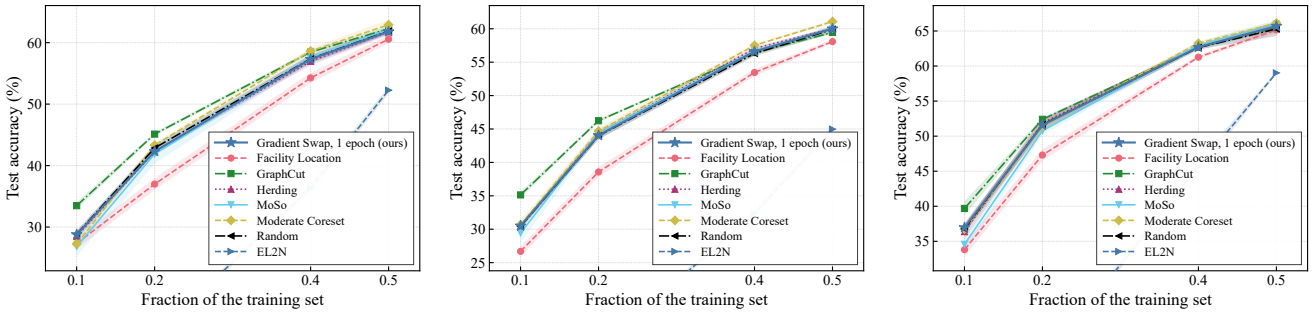
Overview of Algorithm 1. Fix a subset I of size m and let $s := \frac{2}{m^2} C\mathbf{1}_I$. Since $\frac{1}{m}\sum_{j \in I} \zeta_j = \nabla L_{S_I} - \nabla L_S$, the entry s_i equals $\frac{2}{m}$ times the trajectory average, weighted as in Equation (8), of $\langle \zeta_i, \nabla L_{S_I} - \nabla L_S \rangle$: it measures how strongly example i is aligned with the current gradient error of the subset. Exchanging $i \in I$ for $j \notin I$, so that $I' = (I \setminus \{i\}) \cup \{j\}$, changes the objective by exactly

$$G_T(I') - G_T(I) = s_j - s_i + \frac{C_{ii} + C_{jj} - 2C_{ij}}{m^2}. \tag{9}$$

The pair (i_k, j_k) minimizes the first-order term $s_j - s_i$ over all $m(n-m)$ exchanges: it drops the selected example most aligned with the error and adds the unselected example most opposed to it. The second term is the weighted squared distance between the gradients of examples i and j along the trajectory. It is nonnegative, so the first-order prediction is optimistic, and the algorithm therefore evaluates the exact objective before accepting the exchange. Every accepted exchange lowers G_T, and with it the faithfulness proxy $\sqrt{G_T(I)/2}$ of Corollary 4.2. Whether the algorithm starts from a uniformly random subset or from the output of another selection method, it returns a subset whose G_T is at least as small as that of its starting point. Note that although G_T is defined along a full-data training run, Section 5 shows that a single early checkpoint suffices in practice.



(a) ResNet-18 (b) VGG-16 (c) Inception



(d) ResNet-18 (e) VGG-16 (f) Inception

Figure 2: Test accuracy of the retrained model versus retention fraction on CIFAR-10 (top row) and CIFAR-100 (bottom row), averaged over five seeds (std in shaded area). Gradient Swap uses $\tau = 1$ (epoch 1). It is competitive with static baselines while being the only method that optimizes the estimated G_T.

Remark 4.5. *Evaluating $v \mapsto Cv$ requires the per-example gradients of all n pool examples at every checkpoint, or an $n \times n$ matrix built at $\Theta(n^2 d)$ cost per checkpoint. Our implementation removes the memory requirement by storing only aggregate gradients.*

Theorem 4.6 (Swap descent decreases the faithfulness bound)**.** *In the setting of Definition 4.4, let $1 \le m \le n-1$. For every initial subset $I_0 \subseteq [n]$ with $|I_0| = m$ and every tie-breaking rule in the choice of (i_k, j_k), Algorithm 1 generates subsets $I_0, I_1, \dots$ of size m and returns I_{out} such that:*

(i) every accepted swap strictly decreases the objective, $G_T(I_{k+1}) < G_T(I_k)$, and the algorithm terminates after finitely many iterations;

(ii) if I_0 is drawn uniformly from the subsets of $[n]$ of size m, then $\mathbb{E}[G_T(I_{\text{out}})] \le \frac{n-m}{m(n-1)} \sum_{t=0}^{T-1} \frac{\beta_t h_t}{4} V_t$, where $V_t := \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}_{\theta \sim \mu_t^S} \|\nabla \ell(\theta, Z_i) - \nabla L_S(\theta)\|^2$ and the expectation is over the initialization.

The proof alongside the construction of the algorithm is detailed in Section F.

Relation to prior work. G_T is not submodular, so greedy combinatorial algorithms and facility-location surrogates (Wei et al., 2015; Iyer et al., 2021) carry no guarantee for it. Swap descent is thus a way to propose swaps from first-order scores but accepts them using the exact quadratic, so accepted swaps never increase G_T (Theorem 4.6).

5 Experiments

We evaluate an SGD-based estimate $\widehat{G}_T$ of G_T and our swap algorithm (Algorithm 1) around three questions: (i) how much of the full-data trajectory is needed to estimate G_T? (ii) does $\widehat{G}_T$ track the quality of the subsets it scores? and (iii) does optimizing $\widehat{G}_T$ via local search match established selectors in accuracy?

5.1 Experimental setup

Datasets and models. We use CIFAR-10 and CIFAR-100 (Krizhevsky, 2009) with three architectures: ResNet-18 (He et al., 2016), VGG-16 (Simonyan and Zisserman, 2015), and Inception (Szegedy et al., 2014). The same datasets and architectures are used for every selection method. Models are trained using SGD with Nesterov momentum 0.9, initial learning rate 0.01, and weight decay 5×10^{-4}. The learning rate follows a cosine schedule for 300 epochs, with batch size 128 and cross-entropy loss. The same optimizer and schedule are used both to train the full-data reference trajectory and to retrain on a selected subset.

Estimating G_T. We do not compute G_T exactly; we estimate it and denote the estimate by $\widehat{G}_T$. We train the full-data reference model once on a fixed random 90%/10% train/validation split and store checkpoints θ_k (caveat in Section G.1). The expectation over μ_t^S in the definition of G_T is replaced by this single realized trajectory, and the sum over steps by a weighted sum over the stored checkpoints. Neither C nor per-example gradients are ever formed. Writing $u_I^k = \sum_{i \in I} \nabla \ell(\theta_k, Z_i)$ and $\bar{g}_k = \nabla L_S(\theta_k)$, we have $\widehat{G}_T(I) = \sum_k w_k \, \|u_I^k/m - \bar{g}_k\|^2$, so only one *aggregate* gradient per checkpoint is stored. The cost is still linear in the number of checkpoints, which makes swaps expensive for long horizons T, since aggregate gradients need to be recomputed after each swap. For selection we therefore use $\tau \in \{1, 4, 10, 15, 20\}$ checkpoints: $\tau = 1$ is a single-checkpoint estimate (epoch 1), and larger τ approaches the full trajectory average of the theory. In Figure 1, $\widehat{G}_T$ is evaluated once per subset with no swap sweeps, so we use *all* 300 stored checkpoints. We refer to Algorithm 1 as Gradient Swap and add a second stopping criterion, a maximum of 200 allowed swaps.

Baselines. We compare against six static data-selection methods: Herding (Welling, 2009), Moderate Coreset (Xia et al., 2022), Facility Location and GraphCut (Iyer et al., 2021), EL2N (Paul et al., 2021), and MoSo (Tan et al., 2023). We also include random selection, which draws a uniform subset. We omit Grad-Match (Killamsetty et al., 2021a) and CRAIG (Mirzasoleiman et al., 2020), since they are *adaptive* methods that reselect the subset periodically during training, whereas we focus on static selection, where a single subset is chosen before training.

Evaluation protocol. Each method selects a subset at four retention fractions of the training pool (10%, 20%, 40%, 50%), and the model is then retrained on the selected subset from scratch. Our main metric is the test accuracy of the retrained model, averaged over five seeds. To assess whether G_T is a meaningful target, we compare the estimated bound $\sqrt{\widehat{G}_T(I)/2}$ to the faithfulness gap $R_P(\mathcal{A}(S_I)) - R_P(\mathcal{A}(S))$ for subsets produced by different selection methods at retention 0.1, where $\mathcal{A}$ is SGD, and report Pearson correlation.

5.2 Results

A single early checkpoint is enough to estimate G_T. Table 1 reports the test accuracy of the selected subsets for $\tau \in \{1, 4, 10, 15, 20\}$. No two settings in a column differ by more than roughly one standard error of their difference, and there is no monotone trend in τ. Additional checkpoints thus buy no detectable improvement while costing proportionally more, so we use $\tau = 1$ in all remaining experiments. This echoes the finding of Paul et al. (2021) that relevant examples can be identified in the early stages of training. Results for other settings are reported in Section J.

Table 1: Test accuracy (%) on CIFAR-10 with ResNet-18 as a function of the number of retained checkpoints τ, at three retention fractions

τ	10%	20%	40%
1	69.57 ± 1.00	80.12 ± 0.80	87.02 ± 0.25
4	69.77 ± 0.85	80.37 ± 0.51	87.05 ± 0.39
10	69.81 ± 1.48	80.13 ± 0.77	87.11 ± 0.34
15	69.06 ± 1.35	80.30 ± 0.80	87.05 ± 0.23
20	69.86 ± 0.72	80.34 ± 0.65	86.97 ± 0.21

$\widehat{G}_T$ separates faithful subsets from unfaithful ones. Figure 1 plots the faithfulness gap against the estimated bound $\sqrt{\widehat{G}_T(I)/2}$. Subsets with large $\widehat{G}_T$ are the ones with the largest gap, with a Pearson correlation of $r = +0.98$ for ResNet-18 on CIFAR-10 and $r = +0.80$ for VGG-16 on CIFAR-100. This holds even though G_T is derived for ULA while the estimate $\widehat{G}_T$ is computed from SGD and the subset models are also retrained with SGD.

Gradient Swap matches established selectors. Figure 2 shows test accuracy against retention fraction. Across datasets and architectures, Gradient Swap performs on par with the static baselines: its accuracy falls within the spread of the baselines at every retention fraction, and it is neither consistently ahead of nor behind them. This is notable because Gradient Swap is a plain local search. We do not read these results as an argument for this particular algorithm. They show that a simple attempt at optimizing the estimate $\widehat{G}_T$ already matches established selectors, which, together with the correlation above, suggests that G_T is a good target for developing stronger selection algorithms. Furthermore, the spread of the selectors in Figure 2 is in line with the DeepCore benchmark (Guo et al., 2022), which concludes that random selection remains a strong baseline. Swap descent can also refine bad selectors: warm-starting it from GraphCut, Herding, Random or EL2N lowers $\widehat{G}_T$ in every case and raises EL2N's accuracy by 2.6 points, while leaving the other three unchanged (Appendix I).

6 Discussion and Limitations

In this paper, we revisited the theoretical and practical foundations of data selection from first principles. We established a fundamental statistical limitation: even when the training sample contains an accurate subset, no selector can reliably identify one from the training sample and a validation set of comparable size (Theorem 2.1). This impossibility result motivated a shift from population risk minimization to *faithfulness*—ensuring subset training closely tracks the full-data parameter distribution. By modeling gradient-based learning via ULA, we bounded the faithfulness gap by the trajectory gradient discrepancy $G_T(I)$, a training-only quantity whose off-diagonal structure captures redundancy. Although minimizing G_T is NP-hard, our swap descent algorithm guarantees a decrease of the bound at every step. Empirically, an efficient single-checkpoint implementation matches established static heuristics on CIFAR-10/100 test accuracy.

While our algorithm performs on par with existing methods on average accuracy, this is highly encouraging: (1) it validates that directly optimizing our mathematically principled proxy (G_T) is sufficient to match top-tier heuristics, and (2) minimizing $G_T(I)$ guarantees faithfulness across *all* bounded downstream queries (Corollary 4.2), providing robustness guarantees where average-case heuristics fail.

Our work opens up many directions for investigation. Our lower bound suggests a deeper connection between data selection and various forms of data memorization (Brown et al., 2021; Attias et al., 2024). Our guarantee is proven for ULA and is one-sided; a faithfulness bound for SGD or Adam would close the gap between analysis and practice. G_T is a difference of submodular functions (Narasimhan and Bilmes, 2012; El Halabi et al., 2023), so stronger solvers apply to the same objective directly, and random

projections (Section K) and proxy models (Coleman et al., 2019) point to selection at scale.

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A Notation

Let $\mathcal{Z}$ denote the space of data points and $P \in \mathcal{P}(\mathcal{Z})$ denote the underlying data distribution. We consider the standard learning problem where we have *i.i.d.* training samples $S = (Z_1, \ldots, Z_n) \sim P^n$ from an unknown data distribution P. For $I \subseteq [n]$, write $S_I := (Z_i)_{i \in I}$. A data selection procedure with budget m identifies an index set $I \subseteq [n]$ with cardinality $|I| \leq m$. Models are parameterized by a vector $\theta \in \mathbb{R}^d$, and the loss incurred on an individual sample Z_i is given by $\ell(\theta, Z_i)$. The full-dataset empirical risk is defined as $L_S(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(\theta, Z_i)$, while the corresponding subset empirical risk is denoted by $L_{S_I}(\theta) = \frac{1}{|I|}\sum_{i \in I} \ell(\theta, Z_i)$. For a given (randomized) learning algorithm $\mathcal{A}$ mapping a dataset to a distribution over $\mathbb{R}^d$, the population risk of a hypothesis $\theta \sim \mathcal{A}(S_I)$ trained on S_I is defined as $R_P(\mathcal{A}(S_I)) := \mathbb{E}_{Z\sim P, \theta \sim \mathcal{A}(S_I)}[\ell(\theta, Z)]$, while the Bayes-optimal population risk is given by $R_P^\star := \inf_f \mathbb{E}_{Z\sim P}[\ell(f, Z)]$, the infimum being over all measurable predictors. For iterative learners, $\mu_t^{S_I}$ denotes the law of the iterate θ_t when training on S_I, and $\mathcal{A}(S_I) = \mu_T^{S_I}$ at the final iteration T.

B Proof of Selection Lower Bound

This appendix proves Theorem 2.1, and is structured as follows: Section B.1 constructs the family $\mathfrak{P}_{\lambda,n,v}$ and a meta-distribution on it. Section B.2 proves the lower bound in Equation (4) by averaging over the meta-distribution, and Section B.3 defines the learner $\mathcal{A}^*$ and proves the upper bounds in Equations (5) and (6) for every member of the family.

The lower bound rests on a comparison of bits. Given the part of the data that does not depend on the labels, a subsample of size m carries at most $n + mB$ bits about the labels, where B is the number of bits in a label: at most n bits for the choice of the subsample and B bits for each selected label. The hard test inputs ask for the labels of a few hidden active edges of a graph. If these edges are unknown, the model

trained on the selected subsample must predict a collection of labels whose entropy exceeds $n + mB$ by order $(1-\lambda)nB$ bits, and each wrong label costs test probability of order $1/(n+v)$. The oracle knows the active edges and can select m examples on which $\mathcal{A}^*$ has expected risk $O_\lambda(n/((n+v)\log n))$, where the expectation is over S.

B.1 Description of the task and the meta-distribution

Each distribution in the family mixes a graph task and a baseline task. In the graph task, each edge of a public cubic graph G carries a label in $\mathbb{F}_2^B$, and the labels of the three edges at every vertex sum to zero. These labels have joint entropy of order NB, where N is the number of edges, but the parity constraints make them locally dependent: if an edge does not appear in the training set and the missing edges around it form a small tree, then the labels of the observed edges around this tree determine its label. This is how full-data training succeeds, and the girth of G is large so that small sets of missing edges are trees. Most test inputs display their own label. The hard test inputs are *masked*: they show an edge without its label, and they ask only about one hidden *active* edge in each group of g edges. Masked examples are rare enough that (S, V) reveals the active edge of only a small fraction of the groups. A selected subsample therefore has to support correct predictions on essentially all edges, whereas the oracle needs only a few examples around each active edge.

The baseline task makes this comparison work for every $\lambda \in (0,1)$. Its labels are given by an unknown matrix Ψ with D rows in $\mathbb{F}_2^B$, where D is slightly smaller than m. A clear baseline example displays a vector q together with the linear combination $q^\top \Psi$ of the rows of Ψ, and a masked baseline input asks for one row. Correct answers to all masked baseline inputs require all of Ψ, that is, DB bits. On the other hand, any D clear baseline examples with linearly independent vectors q determine Ψ, and the training set contains D such examples with high probability. Separate subpopulations with independent labels would not have this property: the training set contains only a constant factor more clear baseline examples than D, so a constant fraction of D separate subpopulations would be absent from it, and even full-data training would fail on them. The baseline task leaves a residual budget of about $\lambda(1-\lambda)n/10$ examples for the graph task. Without it, the argument would need the graph labels alone to carry more than $n + mB$ bits, but their entropy is about $NB/3 < nB/3$, which is below mB once $\lambda > 1/3$.

We now give a concrete choice of parameters. Write $\gamma := 1-\lambda$ and $m := \lfloor \lambda n \rfloor$, and set

$$B := \left\lceil \frac{100}{\gamma} \right\rceil, \quad K := 2^B, \quad g := \left\lceil \frac{100}{\lambda} \right\rceil, \quad \kappa := \lambda - \frac{\lambda\gamma}{10}, \quad D := \lfloor \kappa n \rfloor, \quad \omega := \frac{\gamma}{100g} \cdot \frac{n}{n+v},$$

and

$$a := \frac{3\gamma}{4}, \qquad p_{\mathrm{c}} := 1 - a - \omega(a+\kappa).$$

Here B is large enough that the n bits describing a subset are small compared with γnB, the groups are large enough that a constant expected number of examples per active edge fits into the residual budget $m - D$, and the factor $n/(n+v)$ in ω keeps the expected number of masked graph examples in (S, V) independent of v. The four kinds of examples defined below have probabilities a, p_{c}, ωa and $\omega\kappa$, which sum to one. Since $\gamma^2 \le \gamma$, we have $a + \kappa = 1 - 7\gamma/20 + \gamma^2/10 \le 1 - \gamma/4$; as $\omega \le \gamma/100$, this gives $p_{\mathrm{c}} - \kappa = (1 - a - \kappa) - \omega(a+\kappa) \ge \gamma/4 - \gamma/100$. In particular, all four probabilities are positive, and

$$a + \kappa \le 1 - \frac{\gamma}{4}, \qquad p_{\mathrm{c}} - \kappa \ge \frac{6\gamma}{25}, \qquad \left| m - D - \frac{\lambda\gamma n}{10} \right| \le 1. \tag{10}$$

Throughout, $n \ge n_\lambda$ for a threshold n_λ that depends only on λ, and we write "for large n" whenever we enlarge it. Since a does not depend on v and Equation (10) holds for every v, none of these thresholds depends on v.

The graph comes from a classical textbook construction: for every integer $L \ge 3$, there is a simple cubic graph with girth at least L and at most $4\sum_{i=1}^{L-2} 2^i < 2^{L+1}$ vertices. The proof uses only the following

properties of G: it is cubic, its girth is at least some L of order $\log n$, and its number of edges N satisfies Equation (11) below. To obtain them, let

$$L := \Big\lfloor \log_2 \frac{\sqrt{n}}{3g} \Big\rfloor,$$

which is at least 3 for large n, and let G_0 be a graph as above; it has fewer than $3 \cdot 2^L \le \sqrt{n}/g$ edges. Let G be the disjoint union of $g\lfloor an/(g|E(G_0)|)\rfloor$ copies of G_0. Then G is cubic, its girth is at least L, and

$$an - \sqrt{n} < N \le an, \qquad g \mid N, \qquad L \ge \tfrac{1}{2}\log_2 n - \log_2(6g). \tag{11}$$

We fix a partition of $E(G)$ into $J := N/g$ groups $\mathcal{G}_1, \dots, \mathcal{G}_J$ of g edges each, and write $e(j,1), \dots, e(j,g)$ for the edges of $\mathcal{G}_j$. The graph G and the partition depend only on (λ, n), and they are public.

The unknown parameters are a triple (Θ, Ψ, h), where

- $\Theta = (\Theta_e)_{e \in E(G)}$, with $\Theta_e \in \mathbb{F}_2^B$, satisfies the parity constraints

$$\Theta_{e_1} + \Theta_{e_2} + \Theta_{e_3} = 0 \quad \text{for every vertex of } G, \text{ where } e_1, e_2, e_3 \text{ are its three edges;} \tag{12}$$

- $\Psi \in \mathbb{F}_2^{D \times B}$;
- $h = (h_1, \dots, h_J) \in [g]^J$, and $e(j, h_j)$ is the active edge of $\mathcal{G}_j$.

We identify the label set $[K]$ with $\mathbb{F}_2^B$, so a label is a single B-bit string, and a wrong string counts as one classification mistake. A subpopulation is either an edge $e \in E(G)$ or a vector $q \in \mathbb{F}_2^D$ (we treat $E(G)$ and $\mathbb{F}_2^D$ as disjoint sets), and its label is

$$y_e := \Theta_e, \qquad y_q := q^\top \Psi.$$

Let $b_1, \dots, b_D$ be the standard basis of $\mathbb{F}_2^D$, so that y_{b_i} is the i-th row of Ψ.

An input is a pair (u, z), where u is a subpopulation and $z \in \mathbb{F}_2^B \cup \{\perp\}$. Under $P_{\Theta,\Psi,h}$, an example (X, Y) has label $Y = y_u$, where its subpopulation u and its input X are drawn as follows:

$$X = \begin{cases} (u, y_u) \ \text{ with } u \text{ uniform on } E(G) & \text{with probability } a, \\ (u, y_u) \ \text{ with } u \text{ uniform on } \mathbb{F}_2^D & \text{with probability } p_{\mathrm{c}}, \\ (e(j, h_j), \perp) \ \text{ with } j \text{ uniform on } [J] & \text{with probability } \omega a, \\ (b_i, \perp) \ \text{ with } i \text{ uniform on } [D] & \text{with probability } \omega\kappa. \end{cases}$$

In the first two cases the example is *clear*: its input displays its label. In the last two it is *masked*: its input shows $\perp$ in place of the label, but the example still carries its label Y. Training, validation and test examples are i.i.d. from $P_{\Theta,\Psi,h}$. The family $\mathfrak{P}_{\lambda,n,v}$ consists of the distributions $P_{\Theta,\Psi,h}$ for all triples as above. It is finite, its number of classes K depends only on λ, and $R^\star_P = 0$ for every P in it, since the label is a function of the input.

For the lower bound, we draw (Θ, Ψ, h) uniformly from the set of all triples; this is the meta-distribution. Equivalently, Θ, Ψ and h are independent, Ψ and h are uniform, and the B coordinates of Θ, viewed as vectors in $\mathbb{F}_2^{E(G)}$, are independent and uniform on the cycle space

$$\mathcal{C}(G) := \{x \in \mathbb{F}_2^{E(G)} : \ x_{e_1} + x_{e_2} + x_{e_3} = 0 \text{ for every vertex of } G, \text{ where } e_1, e_2, e_3 \text{ are its three edges}\}.$$

This space is defined by $|V(G)| = 2N/3$ linear equations in N unknowns, so $\dim \mathcal{C}(G) \ge N/3$. Measuring entropy in bits, we get

$$H(\Theta) \ge \frac{NB}{3} \qquad \text{and} \qquad H(\Psi) = DB. \tag{13}$$

B.2 Lower bound

Fix a learner $\mathcal{A}_m$ and a selector Π. Throughout this subsection, (Θ, Ψ, h) is drawn from the meta-distribution, then $S \sim P^n_{\Theta,\Psi,h}$ and $V \sim P^v_{\Theta,\Psi,h}$, and Π and $\mathcal{A}_m$ use internal randomness that is independent of everything else; $\mathbb{P}$, $\mathbb{E}$ and H refer to all of this randomness. Let $\widehat{I}_m = \Pi(S, V)$, and let $\hat{f}$ be the classifier output by $\mathcal{A}_m$ on $S_{\widehat{I}_m}$. We will show that

$$\mathbb{E}[R_{P_{\Theta,\Psi,h}}(\hat{f})] \geq \frac{\lambda(1-\lambda)^2}{10^6} \cdot \frac{n}{n+v}. \tag{14}$$

The two steps of the proof, Lemmas B.1 and B.2 below, condition on the part of the data that does not depend on the labels. Let W consist of the subpopulations of the $n+v$ examples in (S, V), the indicators of which of them are masked, and the internal randomness of Π and $\mathcal{A}_m$. The inputs themselves are not part of W, because a clear input contains its label. Given W, an example with subpopulation u equals $((u, y_u), y_u)$ if it is clear and $((u, \perp), y_u)$ if it is masked, so (S, V), $\widehat{I}_m$ and $\hat{f}$ are functions of (W, Θ, Ψ). The subpopulations and the clear/masked indicators are drawn without looking at the labels, so W is independent of (Θ, Ψ). It is not independent of h, since a masked graph example reveals the active edge of its group. Given W, the selected sample is determined by its indices and labels, even though its inputs may be long.

The selector cannot know the active edge of a group from which (S, V) contains no masked example, so we count mistakes on all edges of such groups. Call a group j *unresolved* if no example in (S, V) is a masked example from $\mathcal{G}_j$. Let $U \subseteq [J]$ be the set of unresolved groups and $E_U := \bigcup_{j \in U} \mathcal{G}_j$; both are functions of W. Let Q be the number of wrong labels that $\hat{f}$ outputs on the masked inputs of all edges of unresolved groups and on the masked baseline inputs:

$$Q := \left|\{e \in E_U : \ \hat{f}(e, \perp) \neq \Theta_e\}\right| + \left|\{i \in [D] : \ \hat{f}(b_i, \perp) \neq b_i^\top \Psi\}\right|.$$

Lemma B.1. $\mathbb{E}[R_{P_{\Theta,\Psi,h}}(\hat{f})] \geq (\omega/n)\,\mathbb{E}[Q]$.

Proof. We first show that, conditionally on (W, Θ, Ψ), the active index h_j of every unresolved group is uniform on $[g]$. Given (Θ, Ψ, h), the $n+v$ examples are independent, and the subpopulation and kind of an example depend on (Θ, Ψ, h) only through h, and only for a masked graph example, whose subpopulation is $e(j', h_{j'})$ for a uniform $j' \in [J]$. Hence the conditional law of W given (Θ, Ψ, h) does not depend on (Θ, Ψ), and it depends on h_j only through the masked examples from $\mathcal{G}_j$, of which there are none when $j \in U$. Since h is uniform on $[g]^J$ and independent of (Θ, Ψ), Bayes' rule shows that h_j is uniform on $[g]$ given (W, Θ, Ψ) for every $j \in U$. Meanwhile, $\hat{f}$ is a function of (W, Θ, Ψ), so it is a fixed classifier under this conditioning.

Under $P_{\Theta,\Psi,h}$, each test input $(e(j, h_j), \perp)$ has probability $\omega a/J$ and each test input $(b_i, \perp)$ has probability $\omega\kappa/D$. Keeping only the masked inputs of unresolved groups and the masked baseline inputs,

$$R_{P_{\Theta,\Psi,h}}(\hat{f}) \geq \sum_{j \in U} \frac{\omega a}{J} \mathbf{1}\{\hat{f}(e(j, h_j), \perp) \neq \Theta_{e(j,h_j)}\} + \sum_{i=1}^{D} \frac{\omega\kappa}{D} \mathbf{1}\{\hat{f}(b_i, \perp) \neq b_i^\top \Psi\}.$$

Take the conditional expectation given (W, Θ, Ψ). Since h_j is uniform on $[g]$ for $j \in U$ and $\hat{f}$ is fixed, the j-th term of the first sum becomes

$$\frac{\omega a}{Jg} \sum_{\ell=1}^{g} \mathbf{1}\{\hat{f}(e(j, \ell), \perp) \neq \Theta_{e(j,\ell)}\}.$$

Since $Jg = N \le an$ and $D \le \kappa n$, both $\omega a/(Jg)$ and $\omega\kappa/D$ are at least ω/n, so

$$\mathbb{E}[R_{P_{\Theta,\Psi,h}}(\hat{f}) \,|\, W, \Theta, \Psi] \ge \frac{\omega}{n} Q.$$

Taking expectations proves the lemma. □

By Lemma B.1, it remains to show that $\hat{f}$ makes many mistakes on the inputs counted by Q. The next lemma compares the entropy of their labels with the information in the selected sample. In its proof, Θ_{E_U} denotes the vector in $(\mathbb{F}_2^B)^{E(G)}$ that agrees with Θ on E_U and is zero elsewhere, which keeps the dimension fixed although E_U is random.

Lemma B.2. *For large n, $\mathbb{E}[Q] \ge D + 7N/30 - m - 2n/B$.*

Proof. We first bound the information in the selected sample. For every $i \in \widehat{I}_m$, W gives the subpopulation and the kind of the i-th training example, and its label Y_i gives the rest. Hence, given W, the selected sample $S_{\widehat{I}_m}$, in the order of increasing indices, is determined by

$$M := (\widehat{I}_m, (Y_i)_{i \in \widehat{I}_m}).$$

The internal randomness of $\mathcal{A}_m$ is part of W, so $\hat{f}$ is a function of (W, M). The set $\widehat{I}_m$ is one of $\binom{n}{m}$ subsets of $[n]$, and each label has B bits, so

$$H(M \mid W) \le \log_2 \binom{n}{m} + mB \le n + mB. \tag{15}$$

The first term cannot be dropped: a selector that sees all labels in S and V can encode information about them in its choice of which examples to keep.

Next, we describe (Θ_{E_U}, Ψ) by the predictions of $\hat{f}$ and a list of corrections. From (W, M), one can compute $\hat{f}$ and its predictions on the inputs $(e, \perp)$, $e \in E_U$, and $(b_i, \perp)$, $i \in [D]$. Let $\chi \in \{0,1\}^{E(G)} \times \{0,1\}^D$ indicate the inputs among these on which $\hat{f}$ errs, with $\chi_e = 0$ for $e \notin E_U$, so that χ has Q ones. A correct prediction at $(e, \perp)$ is Θ_e, and a correct prediction at $(b_i, \perp)$ is the i-th row of Ψ. Therefore, given (W, M, χ), every coordinate of (Θ_{E_U}, Ψ) is known except the Q labels at the positions where χ equals one, and (Θ_{E_U}, Ψ) takes at most 2^{BQ} values. By the chain rule,

$$H(\Theta_{E_U}, \Psi \mid W, M) \le H(\chi \mid W, M) + H(\Theta_{E_U}, \Psi \mid W, M, \chi) \le N + D + B\,\mathbb{E}[Q] \le n + B\,\mathbb{E}[Q], \tag{16}$$

where we used that χ takes at most 2^{N+D} values, and in the last step that $N + D \le (a + \kappa)n < n$ by Equations (10) and (11).

Finally, we bound the same entropy from below. Since $H(\Theta_{E_U}, \Psi \mid W, M) \ge H(\Theta_{E_U}, \Psi \mid W) - H(M \mid W)$, it suffices to bound $H(\Theta_{E_U}, \Psi \mid W)$. The labels of different edges are not independent, so we start from the entropy of all of Θ. The matrix Ψ is independent of (Θ, W), and Θ is independent of W. Moreover, Θ is determined by Θ_{E_U} together with its restriction to $E(G) \setminus E_U$, which takes at most $2^{B|E(G)\setminus E_U|}$ values given W. Hence

$$H(\Theta_{E_U}, \Psi \mid W) = H(\Psi) + H(\Theta_{E_U} \mid W) \ge DB + H(\Theta) - B\,\mathbb{E}\,|E(G) \setminus E_U|.$$

To bound the last term, note that an example is a masked example from $\mathcal{G}_j$ with probability $\omega a/J = \omega a g/N$. Since $(n + v)\omega = \gamma n/(100g)$ and $an/N \le 2$ for large n by Equation (11), a union bound gives

$$\mathbb{P}(j \notin U) \le \frac{(n+v)\omega a g}{N} = \frac{\gamma}{100} \cdot \frac{an}{N} \le \frac{1}{10},$$

so $\mathbb{E}\,|E(G) \setminus E_U| = g \sum_{j=1}^{J} \mathbb{P}(j \notin U) \le N/10$. Together with Equation (13), this gives

$$H(\Theta_{E_U}, \Psi \mid W) \ge \Big(D + \frac{N}{3} - \frac{N}{10}\Big)B = \Big(D + \frac{7N}{30}\Big)B. \tag{17}$$

Combining Equations (15) to (17) yields $n + B\,\mathbb{E}[Q] \ge (D + 7N/30)B - n - mB$, and dividing by B proves the lemma. □

Proof of Equation (4). By the definitions and Equation (11), we have $D \ge \kappa n - 1$, $N \ge an - \sqrt{n}$ with $a = 3\gamma/4$, $m \le \lambda n$, $\lambda - \kappa = \lambda\gamma/10 \le \gamma/10$ and $2/B \le \gamma/50$. Hence, by Lemma B.2, for large n,

$$\mathbb{E}[Q] \ge \Big(-\frac{\gamma}{10} + \frac{7}{30} \cdot \frac{3\gamma}{4} - \frac{\gamma}{50}\Big)n - 1 - \sqrt{n} = \frac{11\gamma}{200}\,n - 1 - \sqrt{n} \ge \frac{\gamma n}{20}.$$

By Lemma B.1 and $g \le 101/\lambda$,

$$\mathbb{E}[R_{P_{\Theta,\Psi,h}}(\hat{f})] \ge \frac{\omega}{n} \cdot \frac{\gamma n}{20} = \frac{\gamma^2}{2000\,g} \cdot \frac{n}{n+v} \ge \frac{\lambda\gamma^2}{202000} \cdot \frac{n}{n+v},$$

which proves Equation (14), since $\gamma = 1 - \lambda$. The left-hand side of Equation (14) is the average, over (Θ, Ψ, h) drawn from the meta-distribution, of $\mathbb{E}_{S,V,\Pi}[R_P(\mathcal{A}_m(S_{\widehat{I}_m}))]$ with $P = P_{\Theta,\Psi,h}$, where the inner expectation includes the internal randomness of Π and $\mathcal{A}_m$. Hence there is a triple, and so a member P of $\mathfrak{P}_{\lambda,n,v}$, for which this inner expectation is at least $\lambda(1-\lambda)^2 n/(10^6(n+v))$. This P may depend on the pair $(\mathcal{A}_m, \Pi)$, whereas the family and $\mathcal{A}^*$ do not. Since $R_P^\star = 0$, the compression error and the selection error in Equation (4) add up to $R_P(\mathcal{A}_m(S_{\widehat{I}_m}))$, which proves Equation (4). □

B.3 A good algorithm exists

We now define the learner $\mathcal{A}^*$ of Theorem 2.1 and prove Equations (5) and (6). In this subsection, $P = P_{\Theta,\Psi,h}$ is an arbitrary fixed member of $\mathfrak{P}_{\lambda,n,v}$, and probabilities are over $S \sim P^n$ only; all bounds are uniform over the family.

The learner $\mathcal{A}^*$ knows the public graph G and the numbers D and B, which depend only on (λ, n); it does not know (Θ, Ψ, h). Given a sample of any size, it ignores the masked examples and forms a linear system over $\mathbb{F}_2$ in the unknowns (Θ, Ψ): the parity equations (12), together with the equation $\Theta_e = z$ for each clear example with input (e, z) and the equation $q^\top \Psi = z$ for each clear example with input (q, z). It computes one solution by Gaussian elimination with a fixed pivoting rule, setting free variables to zero (if there is no solution, it takes $\Theta = 0$ and $\Psi = 0$). On a clear input (u, z), it outputs z; on a masked input $(u, \perp)$, it outputs the value of y_u in the computed solution.

Since $\mathcal{A}^*$ copies the label displayed in a clear input, it is correct on every clear input whatever its training sample, so its risk is always at most the total probability $\omega(a + \kappa)$ of the masked inputs. For masked inputs, we say that a subsample of S *determines* y_u if all solutions of its linear system have the same value of y_u. The true (Θ, Ψ) satisfies every equation coming from S, so $\mathcal{A}^*$ is correct on $(u, \perp)$ whenever its training subsample determines y_u. A larger subsample of S only adds equations that the true (Θ, Ψ) satisfies, which cannot enlarge the set of solutions. Hence, if a subsample of S determines y_u, so does every subsample of S that contains it.

To control the graph labels, we look at the edges that do not appear in S. Call an edge *observed* if S contains a clear example of it, and *missing* otherwise. For a missing edge e, let T_e be the connected component containing e of the subgraph of G formed by the missing edges, and write $|T_e|$ for its number of edges; for an observed edge e, let $T_e := \emptyset$. For a component T of missing edges, let ∂T be the set of

edges outside T that share a vertex with T; these edges are observed, since a missing edge sharing a vertex with T would belong to T. We cut components off at

$$s := \lfloor L/2 \rfloor,$$

which satisfies $(\log_2 n)/8 \le s \le \log_2 n$ for large n by Equation (11). Since $s < L$, a component with at most s edges contains no cycle, so it is a tree. The *certificate* of an edge e is the set $I_e \subseteq [n]$ defined as follows:

- if e is observed, I_e consists of the smallest index of a clear example of e;
- if e is missing and $|T_e| \le s$, I_e consists of the smallest indices of clear examples of the edges of ∂T_e, one index per edge;
- if e is missing and $|T_e| > s$, then $I_e := \emptyset$, and we say that the certificate of e *fails*.

The next lemma collects the properties of certificates that are used in both upper bounds.

Lemma B.3. *For every edge e and every $P \in \mathfrak{P}_{\lambda,n,v}$, the following hold.*

(i) If the certificate of e does not fail, then every subsample of S that contains S_{I_e} determines Θ_e.

(ii) $|I_e| \le s + 3$ and $\mathbb{E}\,|I_e| \le 5$.

(iii) The certificate of e fails with probability at most $3/s$.

Proof. (i) If e is observed, the subsample contains the equation $\Theta_e = y_e$. Otherwise, T_e is a tree, and removing e splits it into two trees. Let V_1 be the vertex set of one of them (possibly a single vertex), and add up the parity equations at the vertices of V_1. Every edge with both endpoints in V_1 appears twice and cancels, so the labels of the edges with exactly one endpoint in V_1 sum to zero. One of these edges is e. Every other edge of T_e has either both or no endpoints in V_1, so the remaining edges of the sum lie in ∂T_e, and the subsample contains a clear example of each of them. Hence its system determines Θ_e as the sum of their labels.

For (ii) and (iii), we first bound the probability that given edges are missing. A training example is a clear example of a fixed edge with probability $a/N \ge 1/n$, by Equation (11). Hence, for any k distinct edges,

$$\mathbb{P}(\text{all } k \text{ edges are missing}) = \Big(1 - \frac{ka}{N}\Big)^n \le \exp(-k) \le \Big(\frac{3}{8}\Big)^k. \tag{18}$$

Next, every edge of T_e is the last edge of a simple path in G that starts with e and consists of missing edges: take a shortest chain of edges of T_e from e to it in which consecutive edges share a vertex; by minimality, edges of the chain that are not consecutive share no vertex, so the chain is a simple path. The path consisting of e alone is the only one with one edge, and since G is cubic, at most 2^ℓ simple paths with $\ell \ge 2$ edges start with e. Applying Equation (18) to the ℓ edges of each path gives

$$\mathbb{E}\,|T_e| \le \frac{3}{8} + \sum_{\ell \ge 2} 2^\ell \Big(\frac{3}{8}\Big)^\ell = \frac{21}{8}. \tag{19}$$

(ii) If e is missing and $|T_e| \le s$, then, since T_e is a tree, it has $|T_e| + 1$ vertices, each of degree three in G, and its edges use $2|T_e|$ of the $3(|T_e| + 1)$ edge-vertex incidences at these vertices. Hence $|I_e| = |\partial T_e| \le |T_e| + 3 \le s + 3$. In every case $|I_e| \le 1 + |T_e| + 3 \cdot \mathbf{1}\{e \text{ missing}\}$, so by Equation (19) and Equation (18) with $k = 1$, we get $\mathbb{E}\,|I_e| \le 1 + 21/8 + 9/8 < 5$.

(iii) The certificate of e fails only if $|T_e| > s$. By Markov's inequality and Equation (19), this has probability at most $21/(8s) \le 3/s$. □

The baseline task is handled by a rank argument.

Lemma B.4. *With probability at least $1 - 2\exp(-\gamma^2 n/32)$, the vectors q of the clear baseline examples in S span $\mathbb{F}_2^D$. On this event, S contains D clear baseline examples whose vectors form a basis of $\mathbb{F}_2^D$, and every subsample of S that contains them determines Ψ.*

Proof. Let $n_c \sim \mathrm{Bin}(n, p_c)$ be the number of clear baseline examples in S. By Equation (10), $p_c n - \gamma n/8 \geq \kappa n + (6/25 - 1/8)\gamma n \geq D + \gamma n/10$, so Hoeffding's inequality gives $\mathbb{P}(n_c < D + \gamma n/10) \leq \exp(-2(\gamma/8)^2 n) = \exp(-\gamma^2 n/32)$. Given n_c, the vectors q of these examples are i.i.d. uniform on $\mathbb{F}_2^D$. They fail to span $\mathbb{F}_2^D$ only if some nonzero $x \in \mathbb{F}_2^D$ satisfies $x^\top q = 0$ for all of them, which for a fixed x has probability 2^{-n_c}. A union bound over the $2^D - 1$ nonzero vectors x shows that, when $n_c \geq D + \gamma n/10$, they fail to span with conditional probability at most $2^{-\gamma n/10} \leq \exp(-\gamma^2 n/32)$. If they span, D of them form a basis. The equations $q^\top \Psi = y_q$ of these D examples determine every row of Ψ, and so do the equations of any larger subsample. □

Proof of Equation (5). The full sample S contains S_{I_e} for every edge e. By Lemma B.3, $\mathcal{A}^*(S)$ is correct on $(e, \perp)$ unless the certificate of e fails, which has probability at most $3/s$. By Lemma B.4, $\mathcal{A}^*(S)$ is correct on all inputs $(b_i, \perp)$ except with probability at most $2\exp(-\gamma^2 n/32)$. The masked graph inputs $(e(j, h_j), \perp)$ have total probability ωa, and the inputs $(b_i, \perp)$ have total probability $\omega\kappa$. Therefore,

$$\mathbb{E}_S[R_P(\mathcal{A}^*(S))] \leq \omega\Big(\frac{3a}{s} + 2\kappa\exp(-\gamma^2 n/32)\Big).$$

Since $\omega(n+v)/n = \gamma/(100g) \leq 1$ and $s \geq (\log_2 n)/8$, multiplying by $(n+v)/n$ gives Equation (5). □

For the oracle bound, it remains to check that the certificates of the active edges fit into the budget. Let

$$F_h := \sum_{j=1}^{J} |I_{e(j,h_j)}|.$$

This is an upper bound on the number of distinct examples in these certificates, since different certificates can share examples.

Lemma B.5. *For large n, $\mathbb{E}F_h \leq 3\lambda\gamma n/80$ and $\mathbb{P}(F_h > m - D) \leq 10^5 s^4/(\lambda^2\gamma^2 n)$.*

Proof. By Lemma B.3(ii), $J = N/g \leq an/g$, $a = 3\gamma/4$ and $g \geq 100/\lambda$,

$$\mathbb{E}F_h \leq 5J \leq 5 \cdot \frac{3\gamma}{4} \cdot \frac{\lambda}{100}\, n = \frac{3\lambda\gamma n}{80}.$$

By Equation (10), $m - D \geq \lambda\gamma n/10 - 1$, so $m - D - \mathbb{E}F_h \geq \lambda\gamma n/16 - 1 \geq \lambda\gamma n/20$ for large n. It remains to control the probability that F_h exceeds the budget. We bound its variance using the Efron–Stein inequality.

We first write F_h as a sum over components. All edges of a component T of missing edges have the same certificate, so

$$F_h = \big|\{j :\ e(j, h_j) \text{ is observed}\}\big| + \sum_{T:\,|T|\leq s} \big|\{j :\ e(j, h_j) \in T\}\big| \cdot |\partial T|,$$

where the sum runs over the components with at most s edges; larger components contribute nothing. Each term of the sum lies in $[0, s(s+3)]$, since T contains at most $|T| \leq s$ active edges and $|\partial T| \leq s + 3$ by the proof of Lemma B.3(ii).

The value of F_h depends on S only through the set of observed edges. Suppose that the status of a single edge f changes, from observed to missing or back, and call a component of missing edges, before or after the change, *affected* if it shares a vertex with f. A component that is not affected before the change is also a component after it, again not affected, with the same active edges and the same set ∂T; the same holds with before and after exchanged. Hence only affected components can change their terms. There are at most two affected components before the change and at most two after: when f is missing, only the component containing f is affected, and when f is observed, at most one component meets each of its two endpoints. The first term of F_h changes by at most one, and the sum by at most $2s(s+3)$, so F_h changes by at most $2s(s+3)+1$. Replacing one example of S by an independent copy changes the status of at most two edges, those of the old and of the new example, and therefore changes F_h by at most $2(2s(s+3)+1) \le 18s^2$. Writing $F_h^{(i)}$ for the value of F_h after the i-th example is replaced, the Efron–Stein inequality gives

$$\operatorname{Var}(F_h) \le \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}[(F_h - F_h^{(i)})^2] \le \frac{n}{2}\,(18s^2)^2 = 162\,n\,s^4.$$

Finally, by Chebyshev's inequality,

$$\mathbb{P}(F_h > m - D) \le \mathbb{P}\Big(F_h - \mathbb{E}F_h > \frac{\lambda\gamma n}{20}\Big) \le \frac{162\,n\,s^4}{(\lambda\gamma n/20)^2} \le \frac{10^5 s^4}{\lambda^2\gamma^2 n}. \qquad \square$$

Proof of Equation (6). Since $I_m^\star$ minimizes $R_P(\mathcal{A}^*(S_I))$ over subsets I of size m, it suffices to exhibit a subset $\widetilde{I} \subseteq [n]$ of size m, which may depend on P as $I_m^\star$ does, and to bound $\mathbb{E}_S[R_P(\mathcal{A}^*(S_{\widetilde{I}}))]$. Call S *good* if its clear baseline examples span $\mathbb{F}_2^D$ and $F_h \le m - D$. If S is good, let $\widetilde{I}$ consist of the indices of D clear baseline examples whose vectors form a basis of $\mathbb{F}_2^D$ and the indices in the certificates $I_{e(1,h_1)}, \dots, I_{e(J,h_J)}$. These are at most $D + F_h \le m$ distinct indices, and we add arbitrary further indices to reach exactly m. If S is not good, let $\widetilde{I} := [m]$.

If S is good, then $S_{\widetilde{I}}$ contains the basis and the certificate of every active edge, so by Lemmas B.3 and B.4 it determines Ψ and the label of every active edge whose certificate does not fail. Hence $\mathcal{A}^*(S_{\widetilde{I}})$ errs only on inputs $(e(j,h_j), \perp)$ whose certificate fails, each of which has probability $\omega a/J$. If S is not good, the risk of $\mathcal{A}^*(S_{\widetilde{I}})$ is still at most $\omega(a+\kappa)$, because $\mathcal{A}^*$ is correct on all clear inputs. By Lemmas B.3 to B.5,

$$\mathbb{E}_S[R_P(\mathcal{A}^*(S_{I_m^\star}))] \le \mathbb{E}_S[R_P(\mathcal{A}^*(S_{\widetilde{I}}))] \le \omega\Big(\frac{3a}{s} + (a+\kappa)\Big(2\exp(-\gamma^2 n/32) + \frac{10^5 s^4}{\lambda^2\gamma^2 n}\Big)\Big).$$

Since $\omega(n+v)/n \le 1$ and $(\log_2 n)/8 \le s \le \log_2 n$, multiplying by $(n+v)/n$ gives Equation (6). The constants and the threshold n_λ depend only on λ, which completes the proof of Theorem 2.1. $\square$

C KL divergence in parameter space upper bounds the faithfulness error

Proof. Let $P = \mathcal{A}_n(S)$ and $Q = \mathcal{A}_m(S_I)$, and write $\delta = \mathbb{E}_P[q] - \mathbb{E}_Q[q]$. If $D_{\mathrm{KL}}(P\|Q) = \infty$, the claim is immediate. Otherwise, set $c = \frac{a+b}{2}$. Since P and Q are both probability measures, $(P-Q)(\mathbb{R}^d) = 0$, so

$$\delta = \int (q - c)\,\mathrm{d}(P - Q).$$

Because q takes values in $[a,b]$, we have $|q - c| \le \frac{b-a}{2}$ pointwise, and therefore

$$|\delta| \le \frac{b-a}{2}\,\|P - Q\|_1 = (b-a)\,\mathrm{TV}(P,Q),$$

where $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2}\|P - Q\|_1$ is the total variation distance. Pinsker's inequality gives

$$\mathrm{TV}(P,Q) \le \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P\|Q)}.$$

Combining the two bounds yields

$$|\delta| \le (b-a)\sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P\|Q)} = \sqrt{\frac{(b-a)^2}{2}\, D_{\mathrm{KL}}(P\|Q)}.$$

The argument applies directly to $|\delta|$, so no separate treatment of the positive and negative parts is needed. This proof applies to every measurable q bounded in $[a,b]$. □

D Proof of the gradient trajectory upper bounding the faithfulness

Proof. For a sample $S' \in \{S, S_I\}$, let $L_{S'}$ be its empirical loss, and let $(\theta_t^{S'})_{t=0}^T$ be the ULA chain

$$\theta_{t+1}^{S'} = \theta_t^{S'} - h_t \nabla L_{S'}(\theta_t^{S'}) + \sqrt{2h_t/\beta_t}\,\xi_{t+1}, \qquad \xi_{t+1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d),$$

started from the same initial law μ_0 for both samples. Write $\mu_t^{S'}$ for the law of $\theta_t^{S'}$, so that $\mathcal{A}_n(S) = \mu_T^S$ and $\mathcal{A}_m(S_I) = \mu_T^{S_I}$, and let $\mathbb{P}^{S'}$ be the law of the path $(\theta_0^{S'}, \dots, \theta_T^{S'})$. Recall that

$$G_T(I) = \sum_{t=0}^{T-1} \frac{\beta_t h_t}{4}\, \mathbb{E}_{\theta_t \sim \mu_t^S} \|\nabla L_{S_I}(\theta_t) - \nabla L_S(\theta_t)\|^2.$$

Since $\ell(\cdot, z)$ is continuously differentiable, the map $\theta \mapsto \nabla L_{S'}(\theta)$ is continuous, so the transition kernels

$$K_t^{S'}(\theta, \cdot) = \mathcal{N}\Big(\theta - h_t \nabla L_{S'}(\theta),\ \tfrac{2h_t}{\beta_t} I_d\Big), \qquad t = 0, \dots, T-1,$$

are measurable. For fixed θ and t, the kernels $K_t^S(\theta, \cdot)$ and $K_t^{S_I}(\theta, \cdot)$ are Gaussians with the same covariance $\frac{2h_t}{\beta_t} I_d$ and means that differ by $h_t(\nabla L_{S_I}(\theta) - \nabla L_S(\theta))$. The KL divergence between two Gaussians with a common covariance Σ is $\frac{1}{2}(\Delta m)^\top \Sigma^{-1} \Delta m$, so

$$D_{\mathrm{KL}}\big(K_t^S(\theta,\cdot) \,\|\, K_t^{S_I}(\theta,\cdot)\big) = \frac{1}{2} \cdot \frac{\beta_t}{2h_t} \cdot h_t^2 \|\nabla L_{S_I}(\theta) - \nabla L_S(\theta)\|^2 = \frac{\beta_t h_t}{4} \|\nabla L_{S_I}(\theta) - \nabla L_S(\theta)\|^2.$$

Both chains are Markov with the same initial law μ_0 and use the same Gaussian noise structure, so the chain rule for KL divergence on Markov paths gives

$$D_{\mathrm{KL}}\big(\mathbb{P}^S \,\|\, \mathbb{P}^{S_I}\big) = \underbrace{D_{\mathrm{KL}}(\mu_0 \| \mu_0)}_{=0} + \sum_{t=0}^{T-1} \mathbb{E}_{\theta_t \sim \mu_t^S} \Big[D_{\mathrm{KL}}\big(K_t^S(\theta_t,\cdot) \,\|\, K_t^{S_I}(\theta_t,\cdot)\big)\Big].$$

The outer expectation is under the marginal μ_t^S of the first path law, which is why G_T involves only the full-data chain. Thus:

$$D_{\mathrm{KL}}\big(\mathbb{P}^S \,\|\, \mathbb{P}^{S_I}\big) = G_T(I).$$

By Jensen's inequality, for every θ, $\|\nabla L_{S'}(\theta)\|^2 \le \frac{1}{|S'|} \sum_{z \in S'} \|\nabla \ell(\theta, z)\|^2$, so

$$\|\nabla L_{S_I}(\theta) - \nabla L_S(\theta)\|^2 \le 2\|\nabla L_{S_I}(\theta)\|^2 + 2\|\nabla L_S(\theta)\|^2 \le \frac{2}{m} \sum_{i \in I} \|\nabla \ell(\theta, Z_i)\|^2 + \frac{2}{n} \sum_{i=1}^n \|\nabla \ell(\theta, Z_i)\|^2.$$

The integrability assumption on $\nabla \ell(\theta, Z_i)$ under μ_t^S, for $t \le T$, therefore ensures that every term of $G_T(I)$ is finite.

The map $(\theta_0, \dots, \theta_T) \mapsto \theta_T$ pushes $\mathbb{P}^S$ and $\mathbb{P}^{S_I}$ forward to μ_T^S and $\mu_T^{S_I}$. Data processing for KL divergence then yields

$$D_{\mathrm{KL}}\big(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I)\big) = D_{\mathrm{KL}}\big(\mu_T^S \,\|\, \mu_T^{S_I}\big) \le D_{\mathrm{KL}}\big(\mathbb{P}^S \,\|\, \mathbb{P}^{S_I}\big) = G_T(I).$$ □

E Proof of the quadratic program formulation of subset selection by optimizing the gradient trajectory

Proof. Write $g_i(\theta) := \nabla \ell(\theta, Z_i)$, $\bar{g}(\theta) := \nabla L_S(\theta) = \frac{1}{n}\sum_{i=1}^n g_i(\theta)$, so that $\zeta_i = g_i - \bar{g}$, and $w_t := \beta_t h_t/4$. Since $\sum_{j=1}^n \zeta_j(\theta) = 0$ for every θ,

$$(C\mathbf{1})_i = \sum_{t=0}^{T-1} w_t \, \mathbb{E}_{\theta\sim\mu_t^S}\Big\langle \zeta_i(\theta), \sum_{j=1}^n \zeta_j(\theta)\Big\rangle = 0.$$

For $v \in \mathbb{R}^n$, linearity gives

$$v^\top C v = \sum_{t=0}^{T-1} w_t \, \mathbb{E}_{\theta\sim\mu_t^S}\Big\|\sum_{i=1}^n v_i \, \zeta_i(\theta)\Big\|^2 \geq 0,$$

so $C \succeq 0$. Finally, for $|I| = m$,

$$\nabla L_{S_I}(\theta) - \nabla L_S(\theta) = \frac{1}{m}\sum_{i\in I} g_i(\theta) - \bar{g}(\theta) = \frac{1}{m}\sum_{i\in I}\zeta_i(\theta),$$

and therefore

$$G_T(I) = \sum_{t=0}^{T-1} w_t \, \mathbb{E}_{\theta\sim\mu_t^S}\Big\|\frac{1}{m}\sum_{i\in I}\zeta_i(\theta)\Big\|^2 = \frac{1}{m^2}\sum_{i,j\in I} C_{ij} = \frac{1}{m^2}\mathbf{1}_I^\top C \mathbf{1}_I.$$

Since $I \mapsto \mathbf{1}_I$ is a bijection from subsets of size m onto $\mathcal{B}_m$, minimizing G_T over such subsets is the stated integer program. □

F Construction and analysis of the Swap algorithm

Setup. Write $w_t := \beta_t h_t/4$. For $\alpha \in \mathcal{B}_m$, let $D(\alpha) = \frac{1}{m^2}\alpha^\top C \alpha$ be the objective of Definition 4.4, so that $D(\mathbf{1}_I) = G_T(I)$, and let

$$s(\alpha) := \nabla D(\alpha) = \frac{2}{m^2} C\alpha, \qquad D(\alpha) = \tfrac{1}{2}\,\alpha^\top s(\alpha).$$

Since $C \succeq 0$ and $C\mathbf{1} = 0$, D is convex and already centered. Fix

$$\alpha^\star \in \arg\min_{\alpha\in\mathcal{B}_m} D(\alpha), \qquad D^\star := D(\alpha^\star), \qquad r := \min\{m, n-m\},$$

and write $[x]_+ := \max\{x, 0\}$. Given an iterate $\alpha_k = \mathbf{1}_{I_k}$ with stored $s_k := s(\alpha_k)$, let

$$i_k \in \arg\max_{i\in I_k} s_k(i), \qquad j_k \in \arg\min_{j\notin I_k} s_k(j), \qquad \gamma_k := s_k(i_k) - s_k(j_k), \qquad \delta_k := e_{j_k} - e_{i_k}.$$

The pair (i_k, j_k) minimizes the linearized change $s_k(j) - s_k(i)$ over all exchanges.

Algorithm 2 Gradient-ranked single-swap descent (indicator form)

Initialize $\alpha_0 := \mathbf{1}_{I_0}$ with $|I_0| = m$, and query $C\alpha_0$ to store s_0.
Compute $D(\alpha_0)$ from $C\alpha_0$ and set $\underline{D}_{-1} := -\infty$.
for $k = 0, 1, \ldots$ **do**
    Choose i_k, j_k, γ_k, and δ_k as above.
    Set $\underline{D}_k := \max\{\underline{D}_{k-1}, D(\alpha_k) - r[\gamma_k]_+\}$ and $\widehat{H}_k := D(\alpha_k) - \underline{D}_k$.
    if $\gamma_k \le 0$ **then**
        return $(\alpha_k, \widehat{H}_k)$.
    Set $\widetilde{\alpha}_k := \alpha_k + \delta_k$, query $C\widetilde{\alpha}_k$, and store $\widetilde{s}_k := s(\widetilde{\alpha}_k)$.
    Compute $\Delta_k := \frac{1}{2}\delta_k^\top(s_k + \widetilde{s}_k)$.
    if $\Delta_k < 0$ **then**
        Set $(\alpha_{k+1}, s_{k+1}) := (\widetilde{\alpha}_k, \widetilde{s}_k)$ and store $D(\alpha_{k+1}) := D(\alpha_k) + \Delta_k$.
    else
        return $(\alpha_k, \widehat{H}_k)$.

This is Algorithm 1 written on indicator vectors, with an early exit and the bookkeeping of $\underline{D}_k$ and $\widehat{H}_k$. Neither changes the iterates, as shown below.

F.1 Proof of the swap gap

Proof. Let $I' = (I_k \setminus \{i_k\}) \cup \{j_k\}$, so that $\widetilde{\alpha}_k = \mathbf{1}_{I'}$. Since D is quadratic,

$$\Delta_k = \tfrac{1}{2}\delta_k^\top(s_k + \widetilde{s}_k) = \delta_k^\top s_k + \frac{1}{m^2}\delta_k^\top C\delta_k = D(\widetilde{\alpha}_k) - D(\alpha_k) = G_T(I') - G_T(I_k),$$

and $\delta_k^\top s_k = -\gamma_k$ with $\delta_k^\top C\delta_k \ge 0$, so $\Delta_k \ge -\gamma_k$. Hence a candidate is accepted if and only if $G_T(I') < G_T(I_k)$. Moreover, if $\gamma_k \le 0$ then $\Delta_k \ge 0$, so the candidate would be rejected anyway. The early exit therefore does not change the iterates, and the algorithm coincides with Algorithm 1.

(i) Since $1 \le m \le n-1$, both I_k and its complement are nonempty, so $i_k \in I_k$ and $j_k \notin I_k$ exist for every tie-breaking rule. Hence $|I'| = m$, and by induction every I_k has size m. Every accepted swap satisfies $G_T(I_{k+1}) < G_T(I_k)$, so the values $G_T(I_0) > G_T(I_1) > \cdots$ are strictly decreasing and the accepted subsets are pairwise distinct. There are only $\binom{n}{m}$ subsets of size m, so at most $\binom{n}{m} - 1$ swaps are accepted. The first rejected candidate (or the early exit) then makes the algorithm return $I_{\text{out}} = I_k$.

(ii) By (i), $G_T(I_{\text{out}}) \le G_T(I_0)$ for every I_0 and every tie-breaking rule, so $\mathbb{E}[G_T(I_{\text{out}})] \le \mathbb{E}[G_T(I_0)]$. For I_0 uniform on the subsets of size m, we have $\mathbb{P}(i \in I_0) = \frac{m}{n}$ and $\mathbb{P}(i, j \in I_0) = \frac{m(m-1)}{n(n-1)}$ for $i \ne j$. Therefore

$$\mathbb{E}[G_T(I_0)] = \frac{1}{m^2}\Big[\frac{m}{n}\operatorname{tr} C + \frac{m(m-1)}{n(n-1)}\sum_{i \ne j} C_{ij}\Big].$$

Since $C\mathbf{1} = 0$, we have $\sum_{i \ne j} C_{ij} = \mathbf{1}^\top C\mathbf{1} - \operatorname{tr} C = -\operatorname{tr} C$, and hence

$$\mathbb{E}[G_T(I_0)] = \frac{\operatorname{tr} C}{m^2} \cdot \frac{m}{n}\Big(1 - \frac{m-1}{n-1}\Big) = \frac{n-m}{m\,n\,(n-1)}\operatorname{tr} C.$$

Finally, with $\zeta_i = \nabla\ell(\cdot, Z_i) - \nabla L_S$,

$$\operatorname{tr} C = \sum_{i=1}^{n} C_{ii} = \sum_{t=0}^{T-1} w_t \sum_{i=1}^{n} \mathbb{E}_{\theta \sim \mu_t^S}\|\zeta_i(\theta)\|^2 = n\sum_{t=0}^{T-1} w_t V_t,$$

so that $\mathbb{E}[G_T(I_{\text{out}})] \le \mathbb{E}[G_T(I_0)] = \frac{n-m}{m(n-1)}\sum_{t=0}^{T-1} w_t V_t$. □

G Experimental details

G.1 Caveat on the train / validation split

The reference trajectory is trained on a single random 90%/10% split (seed 42), whereas the selection experiments use five seeds, each with its own split. Only one seed therefore shares its pool with the reference run. For the other four, the selection pool S_s and the reference pool S_{ref} are different 45,000-example draws from the same 50,000 images, overlapping on about 90% of their elements. For these seeds, both the checkpoints used to compute $\widehat{G}_T$ and the full-data model $\mathcal{A}(S)$ in the faithfulness gap of Figure 1 come from S_{ref} rather than S_s. We did not retrain a reference trajectory per seed because of the cost (5× the most expensive step).

G.2 More empirical details

For each (dataset, architecture) pair we train one full-data model on the pool of seed 42 with the protocol of Section 5, and store the parameters at the end of selected epochs, together with the step index and learning rate η_k of that epoch. The *same* reference trajectory is used for all five seeds (see caveat in Section G.1).

We use the SGD objective: $w_k = \eta_k / \sum_j \eta_j$, where η_k is the cosine-schedule learning rate of the epoch e_k in which checkpoint k was saved, $\eta_k = \frac{1}{2}\eta_0(1 + \cos(\pi(e_k - 1)/300))$ with $\eta_0 = 0.01$. The weights are normalized to sum to one, so $\widehat{G}_T$ is a weighted mean over checkpoints. This rescales $\widehat{G}_T$ but changes no swap decision.

Checkpoints used for selection. For τ checkpoints we take the stored epochs closest to τ points equally spaced in optimizer-step time between epoch 1 and epoch 300. Hence $\tau = 1$ gives the epoch-1 checkpoint, and $\tau = 4$ gives approximately epochs $1, 100, 200, 300$. Baseline comparison experiments (Figure 2) use $\tau = 1$; Section J varies τ.

H Empirical correlation between faithfulness error and trajectory gradient discrepancy

Below, we provide results for other settings showing the correlation between the faithfulness error and the estimate of the proxy, $\sqrt{\widehat{G}_T(I)}$. We use similar settings as those explained in Section 5 and Section G. The estimate of G_T is computed using a single training run on $45,000$ sampled points from the training dataset.

Reference trajectory. For each of the six (dataset, architecture) pairs, we train one model on the 45,000 sampled training points using SGD, batch size 128, initial learning rate 0.01 annealed by a cosine schedule over 300 epochs, and we store the parameters θ_k. This run plays the role of $\mathcal{A}(S)$ and is the only training on S that the proxy needs.

Estimating the discrepancy. For a candidate subset I we evaluate

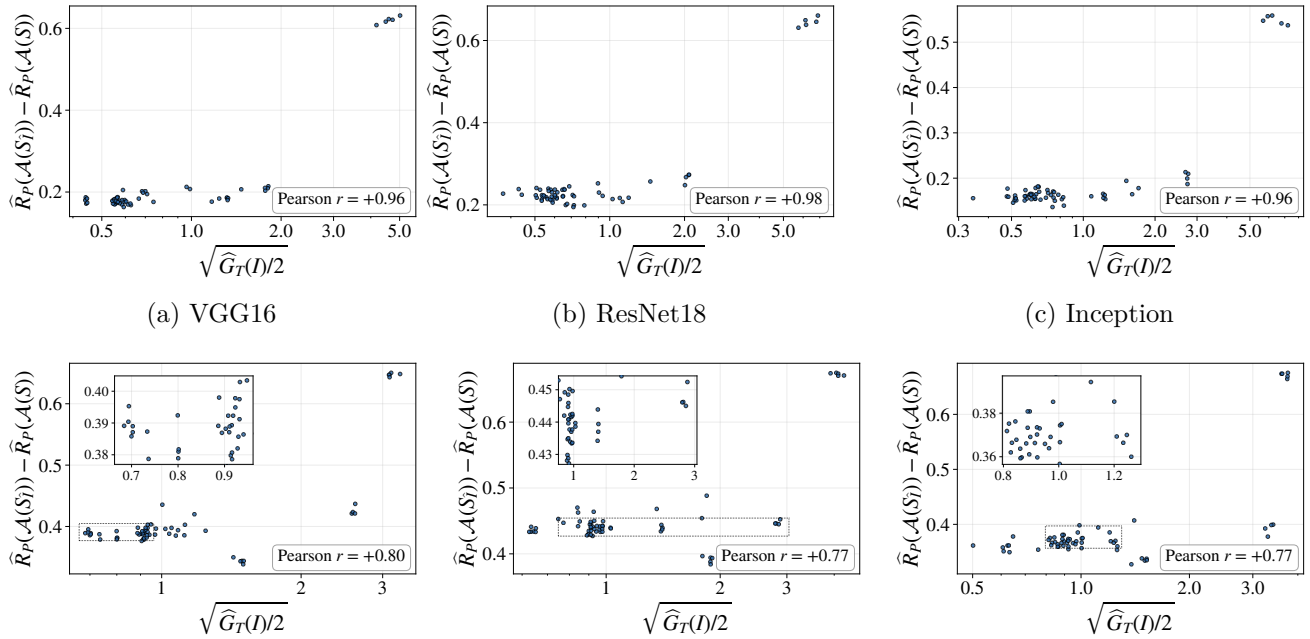
$$\widehat{G}_T(I) \;=\; \sum_k w_k \left\| \nabla L_{S_I}(\theta_k) - \nabla L_S(\theta_k) \right\|_2^2,$$

the sum running over the stored checkpoints, with weights $w_k \propto \eta_k$ normalised to sum to one. The two mean gradients are computed exactly, over *all* trainable parameters and over the whole pool. Examples are viewed through the deterministic test transform and the network is in evaluation mode, so each

per-example gradient is a well-defined function of θ_k alone. This replaces the expectation over μ_t^S by its value at a single realised trajectory.

Candidate subsets and faithfulness error. The points of each panel are the subsets returned by the selection methods of Section 5, including swap-descent variant, at retention fraction 0.1 and over the 5 seeds; one point is one (method, seed) pair, with no averaging. Every subset is retrained from a shared initialization with the same optimizer, schedule and number of epochs as the reference run, and the official test set is evaluated once, after the final epoch. Each panel fixes the dataset, the architecture *and* the selection ratio is set to 10%.

Results. Figure 3 reports the Pearson correlation between the estimated faithfulness and the estimated $\widehat{G}_T$, pooled over methods and seeds within each panel. It exceeds 0.70 in all six settings, so across architectures and datasets the subsets with the largest trajectory gradient discrepancy are also the ones whose retrained model departs most from full-data training.



(a) VGG16 (b) ResNet18 (c) Inception

(d) VGG16 (e) ResNet18 (f) Inception

Figure 3: Correlation between the empirical faithfulness gap and the estimate of the proxy $\sqrt{\widehat{G}_T(I)/2}$. Consistent correlation is shown across datasets (CIFAR10, first row, CIFAR100 second row) and architecture. The zoomed insets show the boxed region.

I Gradient-Swap with warm starts

In order to show the benefit of minimizing the proxy $\widehat{G}_T$ as a way to improve the faithfulness of a subset, we ask whether, starting from a subset obtained by a given data selection method—we pick GraphCut (one of the best-performing baselines), Random, Herding, and EL2N (one of the least-performing ones)—applying swaps of Gradient Swap improves both the $\widehat{G}_T$ objective and downstream accuracy. As expected from Theorem 4.6, the swaps lower $\widehat{G}_T$ for every starting selector, even though τ=10 checkpoints are used for the swaps and all 300 are used for evaluation. The downstream accuracy results reflect the one-sided nature of Corollary 4.2: once a selector already lies inside the low-$\widehat{G}_T$ faithfulness envelope

Table 2: Warm start of full-network Gradient Swap on cifar10 / resnet18 at 10% of the pool. Accuracy is the last logged test epoch. G_T is the full-network objective on every reference checkpoint. The after columns are exact full-network Gradient Swap at $\tau = 10$ started from the baseline subset. Mean and standard deviation are over 5 seeds.

Method	Acc. before	Acc. after	G_T before	G_T after
GraphCut	72.60 ± 0.54	72.16 ± 0.60	3.414 ± 0.2469	3.13 ± 0.2379
Herding	71.74 ± 1.56	71.56 ± 0.84	0.8044 ± 0.1356	0.7549 ± 0.1253
Random	71.66 ± 0.69	72.07 ± 0.11	0.7889 ± 0.2079	0.6516 ± 0.1152
EL2N	29.66 ± 1.12	**32.29 ± 1.11**	81.22 ± 4.636	**68.61 ± 3.138**

Table 3: Test accuracy (%) of full-network Gradient Swap on CIFAR10. Values are mean ± standard deviation over seeds.

Architecture	τ	Test accuracy			
		10%	20%	40%	50%
ResNet-18	1	69.57 ± 1.00	80.12 ± 0.80	87.02 ± 0.25	88.86 ± 0.30
ResNet-18	4	69.77 ± 0.85	80.37 ± 0.51	87.05 ± 0.39	88.91 ± 0.18
ResNet-18	10	69.81 ± 1.48	80.13 ± 0.77	87.11 ± 0.34	88.69 ± 0.31
ResNet-18	15	69.06 ± 1.35	80.30 ± 0.80	87.05 ± 0.23	88.64 ± 0.23
ResNet-18	20	69.86 ± 0.72	80.34 ± 0.65	86.97 ± 0.21	88.73 ± 0.22
VGG-16	1	73.34 ± 0.71	80.86 ± 0.55	86.31 ± 0.26	87.92 ± 0.22
VGG-16	4	72.74 ± 0.78	80.57 ± 0.47	86.34 ± 0.36	87.68 ± 0.23
VGG-16	10	73.41 ± 0.53	80.63 ± 0.25	86.19 ± 0.13	87.68 ± 0.25
VGG-16	15	72.84 ± 0.85	80.72 ± 0.58	86.22 ± 0.29	87.91 ± 0.23
VGG-16	20	72.40 ± 0.94	80.74 ± 0.41	86.39 ± 0.44	87.79 ± 0.28
Inception	1	75.34 ± 1.11	83.55 ± 0.31	88.43 ± 0.21	89.46 ± 0.24
Inception	4	75.22 ± 1.13	83.71 ± 0.23	88.10 ± 0.57	89.33 ± 0.34
Inception	10	75.35 ± 0.78	83.75 ± 0.30	88.27 ± 0.29	89.43 ± 0.22
Inception	15	75.00 ± 0.80	83.62 ± 0.44	88.47 ± 0.23	89.61 ± 0.29
Inception	20	75.93 ± 0.49	83.61 ± 0.25	88.44 ± 0.31	89.39 ± 0.35

(GraphCut, Herding, and Random), the upper bound is already small and further reducing $\widehat{G}_T$ leaves test accuracy unchanged within one standard deviation. Conversely, for EL2N, whose subsets severely violate trajectory fidelity ($\widehat{G}_T = 81.22$), just K=500 swaps reduce $\widehat{G}_T$ by 16% and improve test accuracy by 2.6 percentage points, although it remains far from the other selectors. This confirms that $\widehat{G}_T$ acts as a one-sided safety certificate: it is most informative for diagnosing and repairing unfaithful subsets, whereas further decreases in the low-discrepancy regime do not translate into additional accuracy gains.

J Ablation study on the number of used checkpoints

Tables 3 and 4 show the results of the checkpoint ablation on CIFAR10 and CIFAR100. Using the same setting as the aforementioned experiments, we can see that the number of used checkpoints does not consistently affect test accuracy for three architectures, motivating the use of $\tau = 1$ as a computational simplification for the estimation of G_T.

Table 4: Test accuracy (%) of full-network Gradient Swap on CIFAR100. Values are mean ± standard deviation over seeds.

Architecture	τ	Test accuracy			
		10%	20%	40%	50%
ResNet-18	1	28.79 ± 0.76	42.31 ± 0.54	57.32 ± 0.76	61.79 ± 0.39
ResNet-18	4	28.71 ± 0.64	42.70 ± 0.55	57.05 ± 0.32	61.77 ± 0.33
ResNet-18	10	28.66 ± 0.44	42.43 ± 0.76	57.18 ± 0.72	61.81 ± 0.28
ResNet-18	15	28.92 ± 0.75	42.49 ± 0.44	57.27 ± 0.80	61.62 ± 0.76
ResNet-18	20	28.79 ± 0.89	42.58 ± 0.48	57.33 ± 0.47	61.92 ± 0.42
VGG-16	1	30.51 ± 0.20	44.04 ± 0.55	56.60 ± 0.52	60.01 ± 0.17
VGG-16	4	30.25 ± 0.93	43.72 ± 0.74	56.37 ± 0.34	60.02 ± 0.37
VGG-16	10	30.74 ± 0.83	43.91 ± 0.51	56.71 ± 0.46	59.75 ± 0.40
VGG-16	15	30.73 ± 0.65	43.99 ± 0.33	56.67 ± 0.56	59.81 ± 0.48
VGG-16	20	30.67 ± 0.38	44.04 ± 0.62	56.65 ± 0.26	59.70 ± 0.13
Inception	1	36.97 ± 0.70	51.58 ± 0.44	62.69 ± 0.43	65.69 ± 0.58
Inception	4	36.78 ± 1.01	51.73 ± 0.92	62.88 ± 0.32	65.73 ± 0.46
Inception	10	36.50 ± 0.91	52.02 ± 0.78	62.65 ± 0.56	65.92 ± 0.37
Inception	15	36.53 ± 1.02	51.84 ± 0.21	62.47 ± 0.48	66.00 ± 0.47
Inception	20	36.71 ± 0.83	51.71 ± 0.88	62.44 ± 0.30	—

K Johnson-Lindenstrauss Embeddings Ablation

Following Xia et al. (2024), we attempt dimensionality reduction techniques (Johnson Lindenstrauss Embeddings (Johnson et al., 1984)) on the gradient to reduce the computational and storage cost of algorithm 1. Full-network Gradient Swap stores one aggregate gradient per checkpoint. A count sketch of width p replaces each aggregate by its image in $\mathbb{R}^p$, and the swap test is exact for that sketched objective. In Tables 5 and 6, we compare $p \in \{1024, 2048, 4096\}$ with the exact aggregates $p = d$ on CIFAR-10 with ResNet-18 ($d \approx 10^7$), at checkpoint counts $\tau \in \{1, 4\}$ and subset fractions 10%, 20%, 40%, and 50%. Each entry is the mean test accuracy and the standard deviation over five seeds. The sketch does not change the selected subset's accuracy. At every fraction and both values of τ, the three sketched runs sit on the exact run: the largest gap is 0.60 points ($\tau = 1$, $p = 2048$, 10%), and that gap is smaller than the seed standard deviation of either run. From 40% upward the gaps are about 0.1 points. Width 1024 is already enough. Increasing p to 2048 or 4096 does not produce a further gain, and the same match holds when the trajectory grows from one checkpoint to four. The embedding can therefore be used in place of the exact aggregates.

Table 5: Test accuracy (%) on CIFAR-10 with ResNet-18 at $\tau = 1$. $p = d$ is the exact aggregate. Mean and standard deviation over five seeds.

Fraction	$p = d$	$p = 1024$	$p = 2048$	$p = 4096$
10%	69.57 ± 1.00	69.50 ± 1.37	68.97 ± 0.76	69.66 ± 0.85
20%	80.12 ± 0.80	80.44 ± 0.54	80.39 ± 0.55	79.94 ± 0.37
40%	87.02 ± 0.25	87.02 ± 0.12	86.96 ± 0.53	87.12 ± 0.16
50%	88.86 ± 0.30	88.74 ± 0.16	88.93 ± 0.29	88.75 ± 0.17

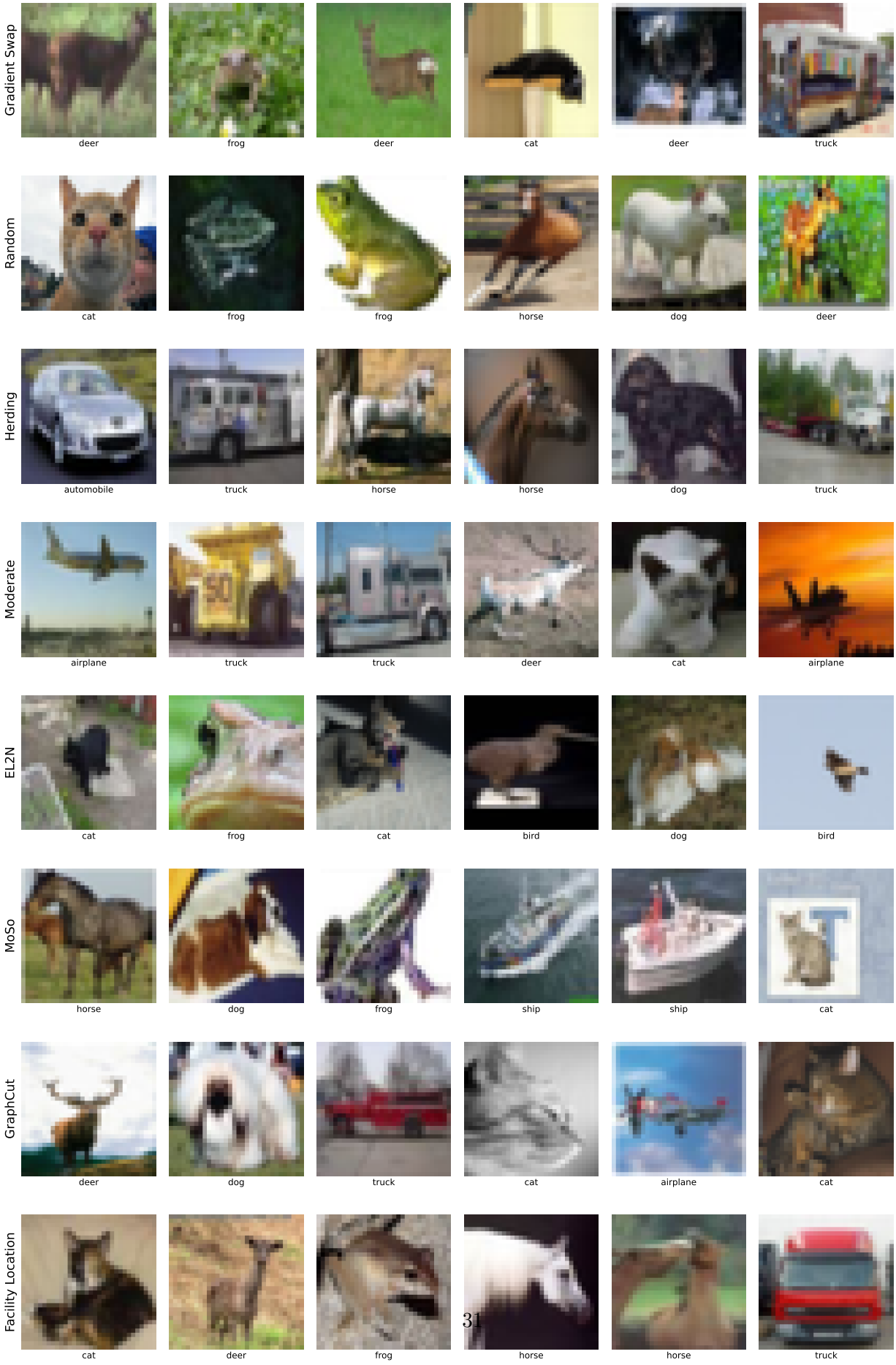
Table 6: Test accuracy (%) on CIFAR-10 with ResNet-18 at $\tau = 4$. $p = d$ is the exact aggregate. Mean and standard deviation over five seeds.

Fraction	$p = d$	$p = 1024$	$p = 2048$	$p = 4096$
10%	69.77 ± 0.85	69.79 ± 0.39	70.04 ± 0.70	69.76 ± 0.65
20%	80.37 ± 0.51	79.93 ± 0.73	80.28 ± 0.47	80.04 ± 0.51
40%	87.05 ± 0.39	87.10 ± 0.22	87.08 ± 0.32	87.09 ± 0.15
50%	88.91 ± 0.18	88.86 ± 0.02	88.84 ± 0.16	88.96 ± 0.14

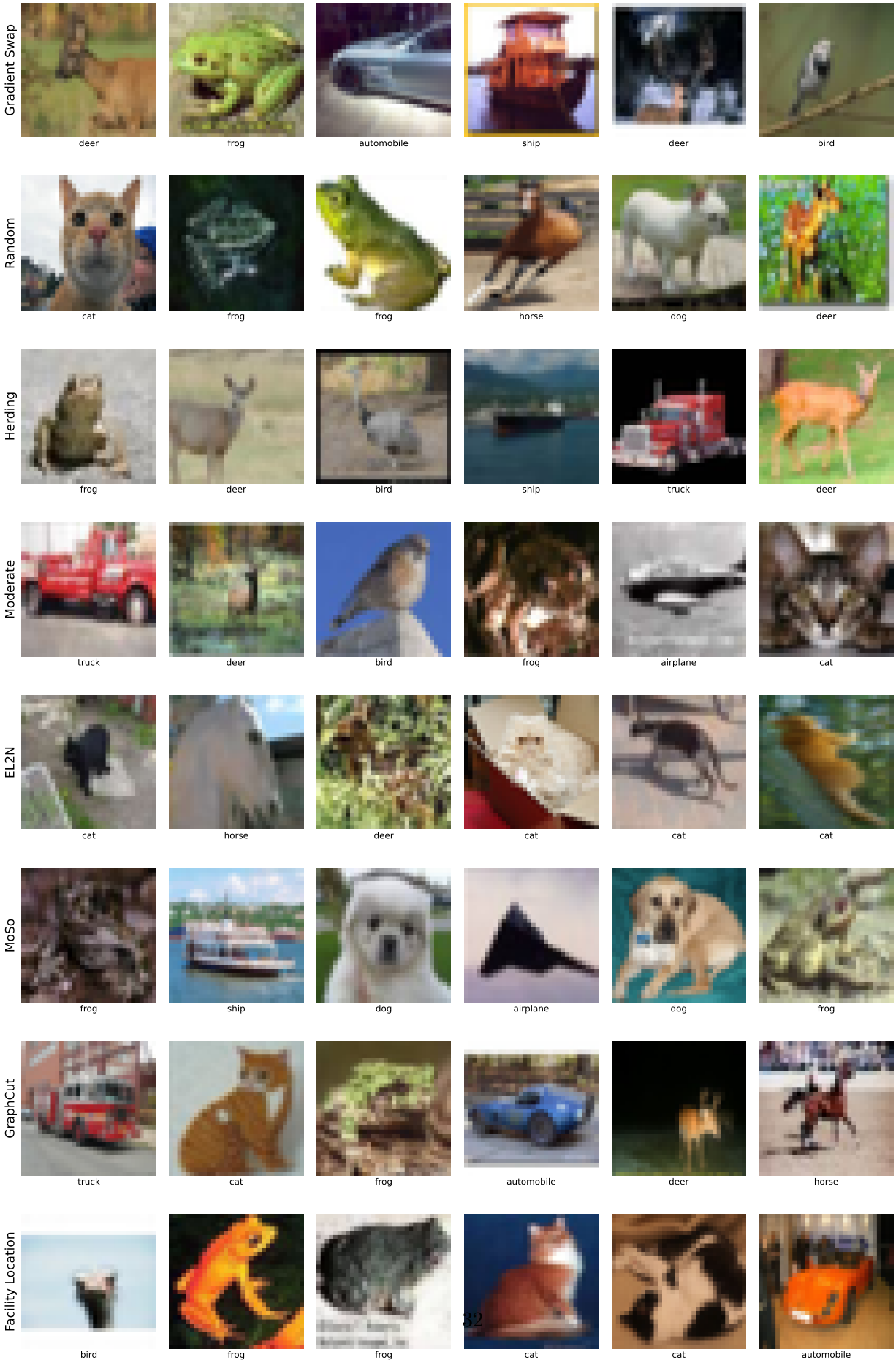
L Selected examples for each selection method

For each dataset and architecture we show the training images selected at 10% of the pool (seed 42). Each row is one method: Gradient Swap using full-network gradients and a single checkpoint ($\tau = 1$), then Random, Herding, Moderate, EL2N, MoSo, GraphCut, and Facility Location. From that method's saved subset we draw six examples without replacement. The stored indices are positions in the 90% training pool; we map them back to the original training set by replaying the same seed-42 split, and we display the raw images with no crop, flip, or normalization. The label under each image is its class.

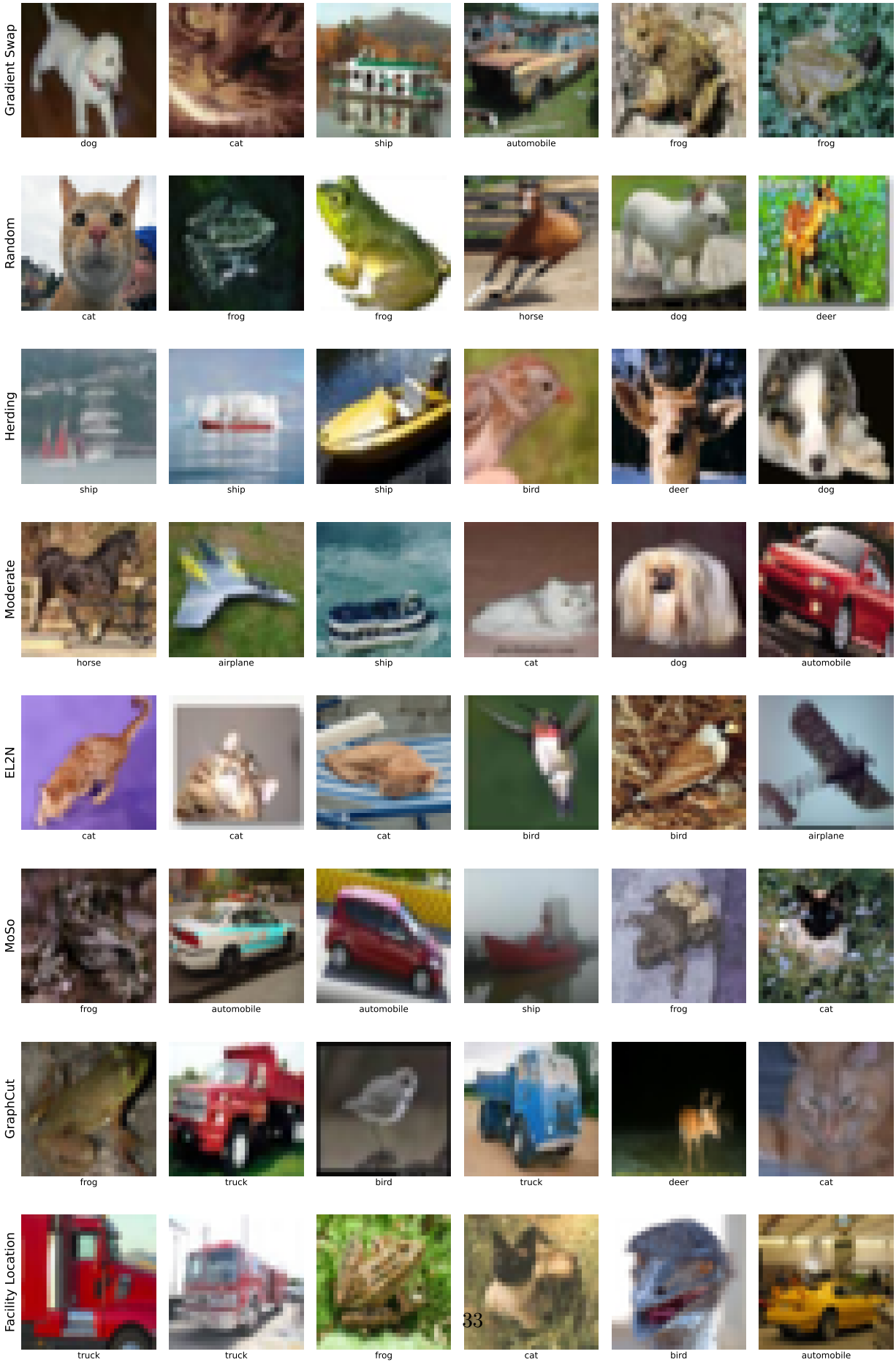
CIFAR-10 / VGG-16, 10% subset, seed 42



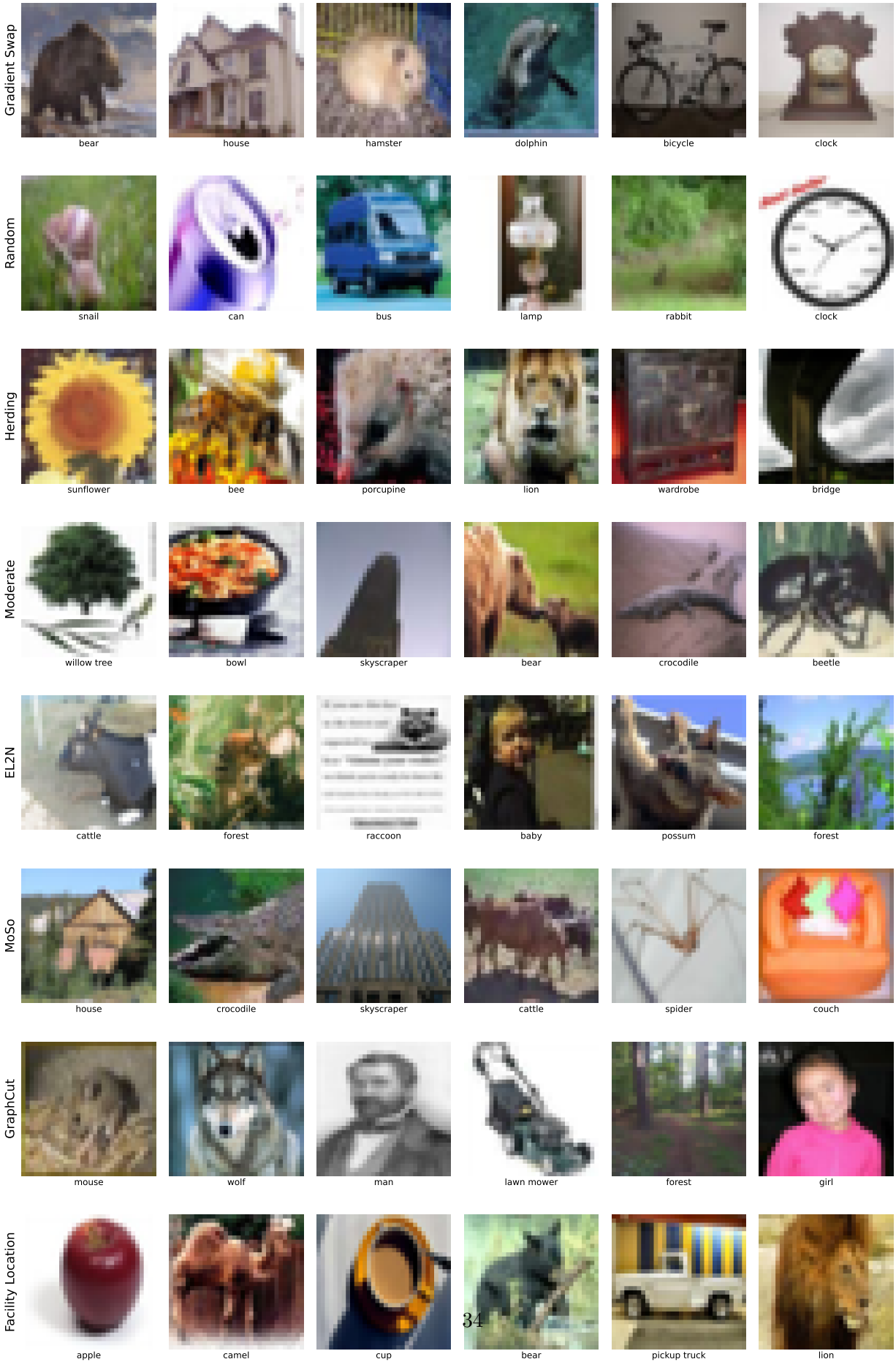
CIFAR-10 / Inception, 10% subset, seed 42



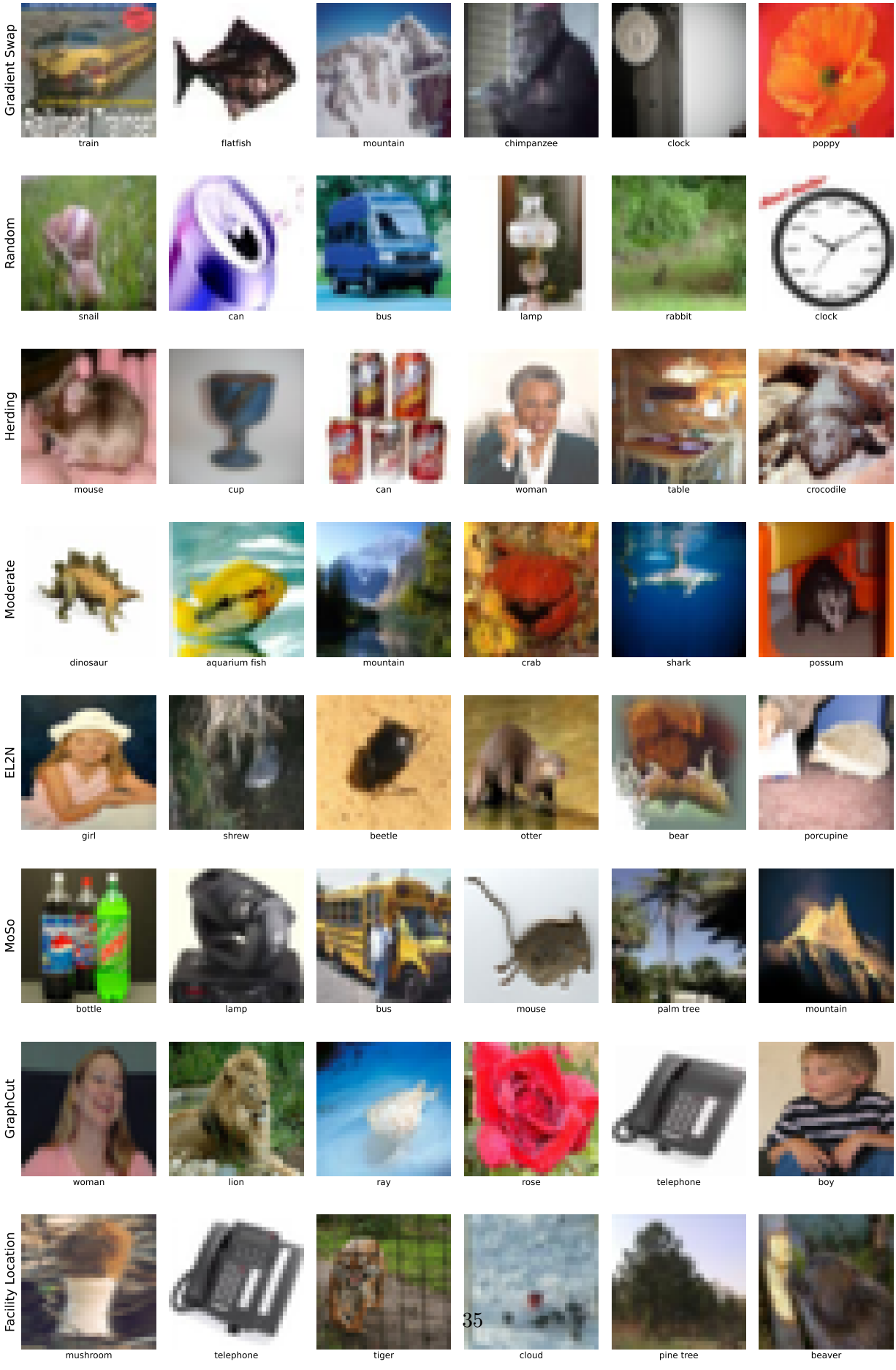
CIFAR-10 / ResNet-18, 10% subset, seed 42



CIFAR-100 / VGG-16, 10% subset, seed 42



CIFAR-100 / Inception, 10% subset, seed 42



CIFAR-100 / ResNet-18, 10% subset, seed 42

